

WELCOME TO SUGARCUBE™

Set-up and Control of the SC2-Mini



Thank You for purchasing the SugarCube SC2-Mini.

Your experience of high-end audio is about to get a lot sweeter.

Instructions for the recording functions are in a separate document: SC2 Recording Guide

For questions, please contact:
support@sweetvinyl.com

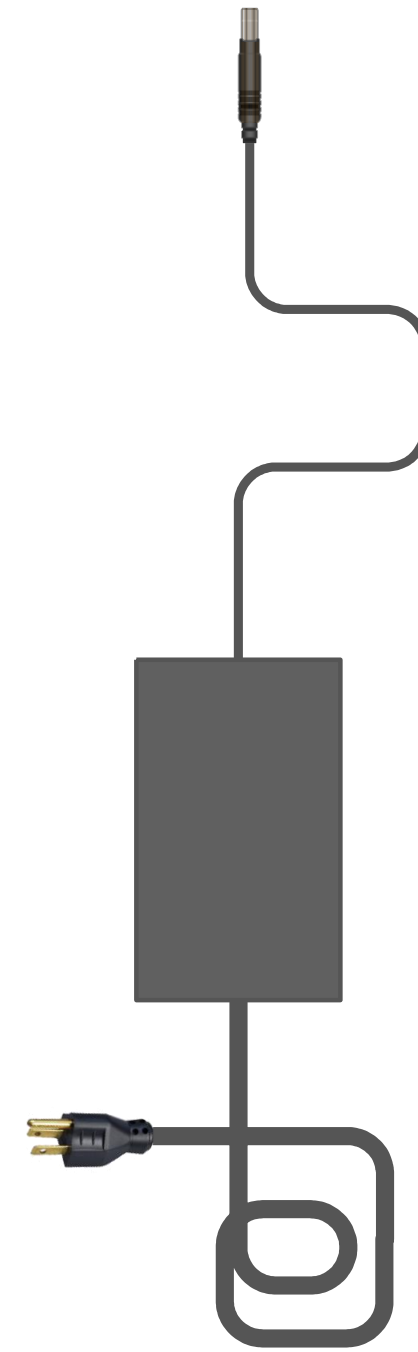


1. Unpacking Your SugarCube



SugarCube SC2-Mini Unit

SugarCube App is available in the mobile app store.



AC Power Adapter

Connects SugarCube to your Wi-Fi network or you can plug an Ethernet cable in the back.



Wi-Fi USB Adapter



USB Stick



Contains the full SugarCube Setup Guide including mobile app and laptop control.

2. The SC2-Mini front panel controls

RECORD

When pressed, SugarCube is recording audio. The **Rec** LED will blink red.

STORAGE

USB thumb drive port accepts USB3 media for transferring recordings. Instructions found in the SC2 Recording Guide.

PAIRING

Press knob to link your device with this SugarCube when asked for a pairing code (Pair).

STATUS DISPLAY

Shows mode of operation, network status, software revision, etc.

CLICK REMOVE

When pressed and the LED is lit, SugarCube is processing audio to remove clicks and pops. The **BYPASS** LED will go out.

BYPASS

When pressed and the LED is lit, an internal relay has shunted the **LINE IN** directly to **LINE OUT**, completely bypassing all internal circuitry.

CLICK MON

When pressed and the LED is lit, the SugarCube suppresses the music so that you can better hear the clicks and pops. This helps set the **STRENGTH** level for particularly noisy LPs.

STRENGTH

Controls the strength of the click and pop detection and removal process. Almost all LPs will sound great at a setting mid-way.

Note: we recommend that you do NOT adjust **STRENGTH** all the way to maximum, you will rarely need this much. Sometimes less is more.



3. The SC2-Mini back panel

DC POWER IN
Connect the supplied adapter to this port.

POWER ON/OFF
Note: when power is off, an internal relay bypasses **LINE IN** to **LINE OUT**.



LINE OUTPUT
Outputs to your line stage (pre-amp). When the front panel **BYPASS** button is pressed, the audio from **LINE IN** passes directly to this **LINE OUT** through a mechanical, audiophile-grade relay.

LINE INPUT
Accepts the output of your phono pre-amp. Please do NOT directly plug your turntable into this port unless it has a built-in phono pre-amp with a line-level output.

WIRED LAN
Accepts standard Ethernet LAN cable

USB PORT FOR WI-FI ADAPTER

Connect either of the above to control the SugarCube from your LAN.

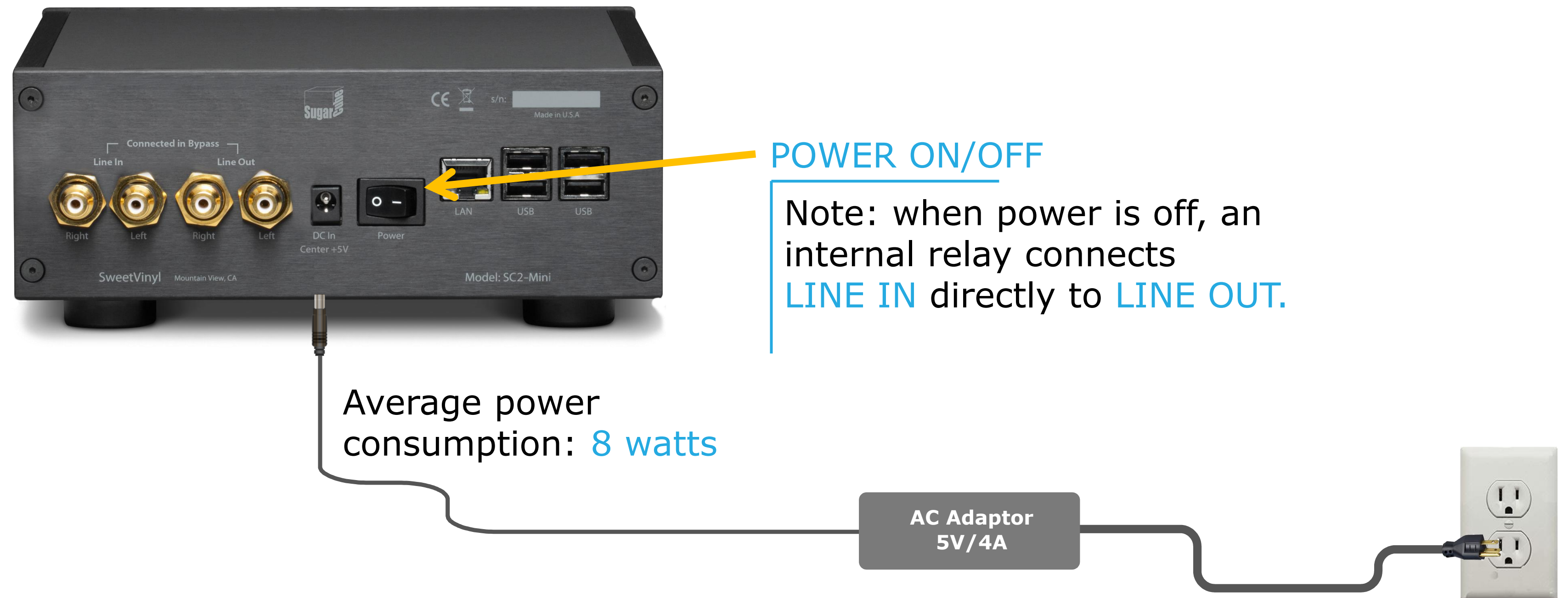
Connecting SugarCube to Your Sound System

1. Hardware Interconnects

PLEASE BE SURE ALL
EQUIPMENT IS TURNED
OFF BEFORE ATTEMPTING
INSTALLATION



4. Connecting Your SC2-Mini



Connecting the supplied AC adaptor.

1. Plug your SugarCube power adapter into a power source.
2. Please only use the supplied 5-volt adaptor or one that we approve.

5. Connecting Your SC2-Mini



Connecting in series between a phono pre-amp and line-stage (pre-amp).

1. Plug your SugarCube power adapter into a power source.
2. Connect the output of your phono pre-amp to the **LINE IN** of the SugarCube.
3. Connect the **LINE OUT** of the SugarCube to an appropriate line-level input of your line-stage/pre-amp.

NOTE: DO NOT CONNECT THE **LINE OUT OF THE SUGARCUBE TO THE PHONO INPUT ON YOUR EQUIPMENT.**

6. Connecting Your SC2-Mini

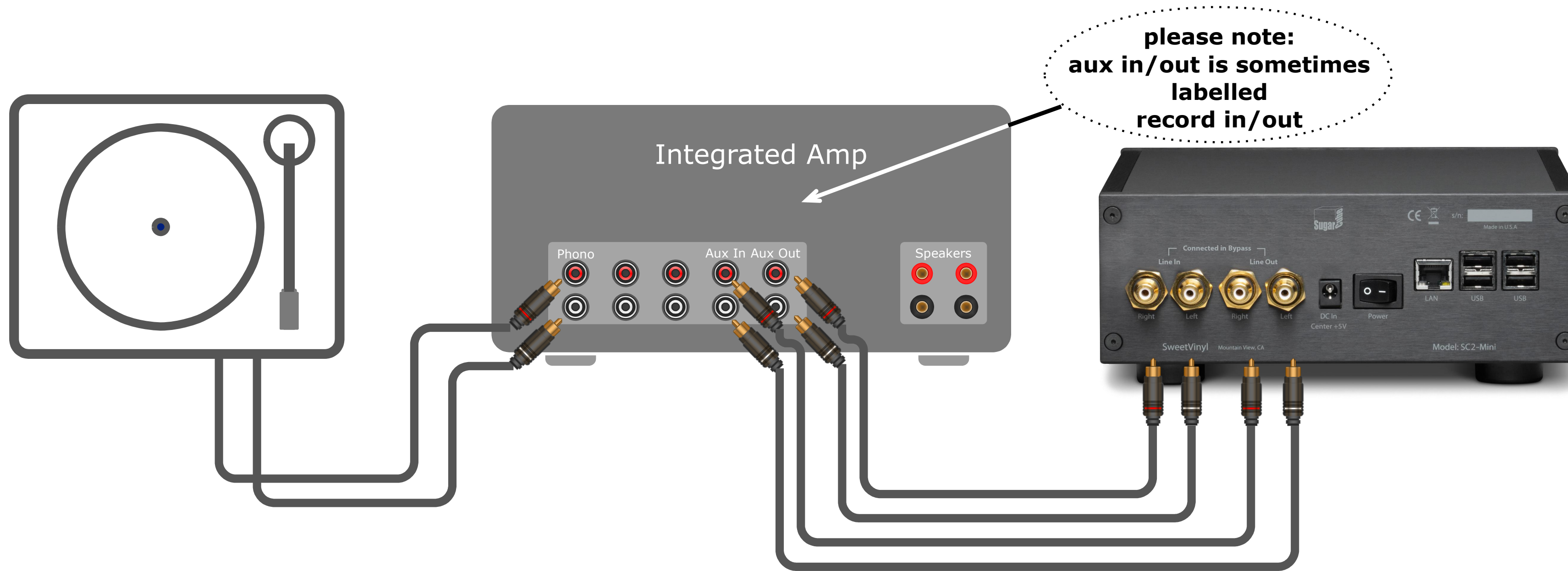


Connecting in series between a pre-amp, with phono input, and power amp

1. Plug your SugarCube power adapter into a power source.
2. Connect the output of your pre-amp to the **LINE IN** of the SugarCube.
3. Connect the **LINE OUT** of the SugarCube to the input of your power amp.

7. Connecting Your SC2-Mini

Integrated Amp: using the 'tape loop'



Connecting to an integrated amplifier via Aux In/Out (or Rec In/Out).

1. Plug your SugarCube power adapter into a power source.
2. Connect **AUX OUT** (or REC OUT) of your integrated amp (or processor) to the **LINE IN** of the SugarCube.
3. Connect **LINE OUT** of the SugarCube to **AUX IN** (or REC IN) of your integrated amp or processor.

NOTE: DO NOT CONNECT THE LINE OUT OF THE SUGARCUBE TO THE PHONO INPUT ON YOUR EQUIPMENT.

8. Connecting Your SC2-Mini

Integrated Amp **WITHOUT** an AUX OUT/IN or REC OUT/IN or PRE-OUT / PWR IN



Many contemporary integrated amplifiers no longer have what was known as a 'tape loop' (back in the day of the cassette or reel-to-reel tape). Also, most modern home theatre processors that do have what are called 'effects loops' will not send audio from a phono input through the loop. In these cases, there is not a way to connect to SugarCube directly to these devices.

The solution is to add a phono stage (pre-amp) between your turntable and the SugarCube. Then connect the Line Out of the SugarCube to any available input on the integrated amp (except for the phono input, of course.)

9. Connecting Your SC2-Mini

Connecting turntables with built-in phono pre-amps



For turntables with a built-in phono pre-amp, be sure to set the **PHONO/LINE** switch to **LINE** before plugging into the SugarCube.

Directly connecting the line-level output from your turntable.

1. Plug your SugarCube power adapter into a power source.
2. Connect the turntable with a built-in phono pre-amp to **LINE IN**.
3. Be sure the turntable is set to **LINE** not **PHONO**.
4. Connect the **LINE OUT** of the SugarCube to any line-level input of your line stage, pre-amp, or integrated amp.

NOTE: SORRY TO BE A BROKEN RECORD ;) BUT DO NOT CONNECT THE LINE OUT OF THE SUGARCUBE TO THE PHONO INPUT ON YOUR EQUIPMENT.

10. Starting Up Your SugarCube



(1) SugarCube is ready to go.



(2) Hold the 'Bypass' button for 5 seconds to see LAN address (and WiFi SSID if using wireless.)

Please note:

When you power up your SugarCube, it will take about [two minutes](#) to complete its startup procedures.

11. Connecting Your SC2-Mini to a LAN or WiFi



Wi-Fi adapter



For controlling your SugarCube, please plug in either the Wi-Fi adapter supplied with your SugarCube OR plug in a wired Ethernet connection. There is no need to connect both except briefly if setting up Wi-Fi (please read appendix A).

NOTE: THE SugarCube's basic functions of click and pop removal are available via front panel controls. However, the recording functions, other than start and stop recording, are only available over a LAN thru a browser or the SugarCube App.

SweetVinyl advises against using a mobile phone for the record functions. We highly recommend using a tablet, laptop, or desktop with recording as there is considerable on-screen detail. (Recording instruction found in a separate document.)

12. Pairing your SC2-Mini

(Please note that pairing is rarely needed)



While setting up the connection between the SugarCube and the SugarCube app, or browser, in rare circumstances you will be prompted to press and release the front panel knob to get a [PAIR](#) code and then enter the four-digit code where directed.

This button may also be used when setting up Wi-Fi for the SugarCube for use with the SugarCube app usually only when there are more than one SugarCube is present.

Controlling SugarCube via a LAN

2. Wired or Wireless

MacOS, Windows
Linux, iOS, or Android



13. Controlling SugarCube via a LAN

Using Wired Ethernet



(1) Type the SugarCube's 'local' address into the **address bar** and press enter.

NOTE: If you own more than one SugarCube, you will need to address the specific SugarCube by its IP address. Press and hold the **Bypass** button for 5 seconds and the IP address will appear on the front panel. Replace 'sugarcube.local:5123' with the IP address appended with ':5123' (example: '191.168.1.45:5123')

15. Controlling SugarCube via a LAN

Setting up WiFi (in place of wired Ethernet), 1 of 4



(1) If wired Ethernet is not available in your sound room, then take the SugarCube to your router just for these Wi-Fi setup steps.

(2) You need both Ethernet and the Wi-Fi stick plugged into the SugarCube simultaneously..

(3) Press Site Survey to find your SSID.

16. Controlling SugarCube via a LAN

Setting up WiFi (in place of wired Ethernet), 2 of 4

State: normal

Ssid: <Not Configured>

Info: disconnected

SSID

Password

Connect

Site survey

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIOT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	BlueCart Wifi	30%	6	WPA2	1
Use	BlueCart Wifi - Guest	50%	6	WPA2	1
Use	SmugMug	84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1
Use	sweetvinyl	100%	9 48	WPA2	4
Use	Tap Guest	34%	11	WPA1 WPA2	1

(4) Please select your network by clicking on the 'Use' button next to your SSID.

Info: disconnected

SSID

Password

Connect

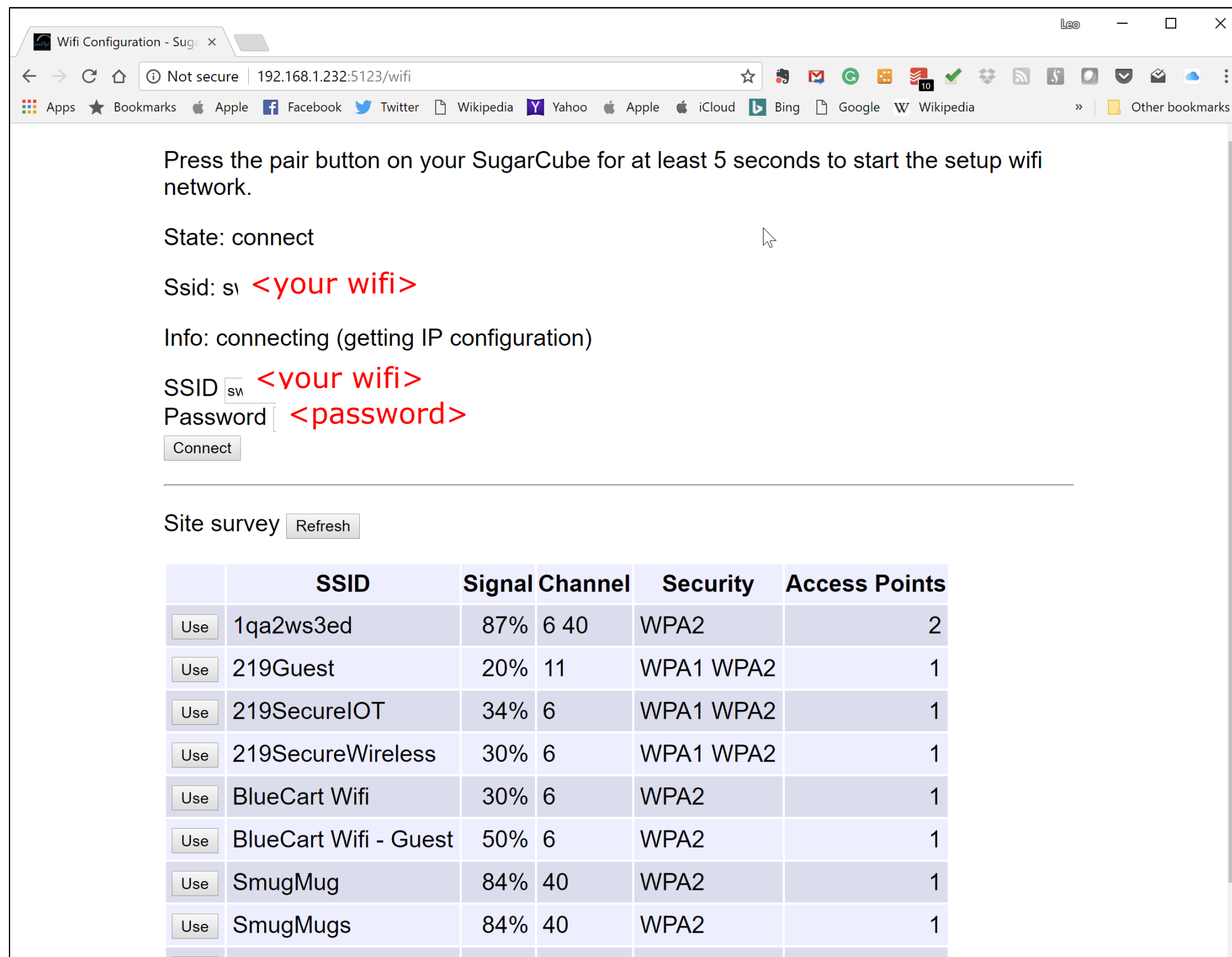
Site survey

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	80%	6 40	WPA2	
Use	219Guest	44%	11	WPA1 WPA2	
Use	219SecureIOT	30%	6	WPA1 WPA2	
Use	219SecureWireless	34%	6	WPA1 WPA2	
Use	BlueCart Wifi	34%	6	WPA2	
Use	SmugMug	87%	6 40	WPA2	

(5) Next, enter the password for your WiFi then click on 'Connect.'

17. Controlling SugarCube via a LAN

Setting up WiFi (in place of wired Ethernet), 3 of 4



Wi-Fi IP address

Your Wi-Fi SSID

Wired IP address



(6) The system will indicate that it is connecting to your Wi-Fi.

(7) Once connected, you will now see both your new Wi-Fi address (w) and your currently wired Ethernet IP address (e).

SugarCube Functions

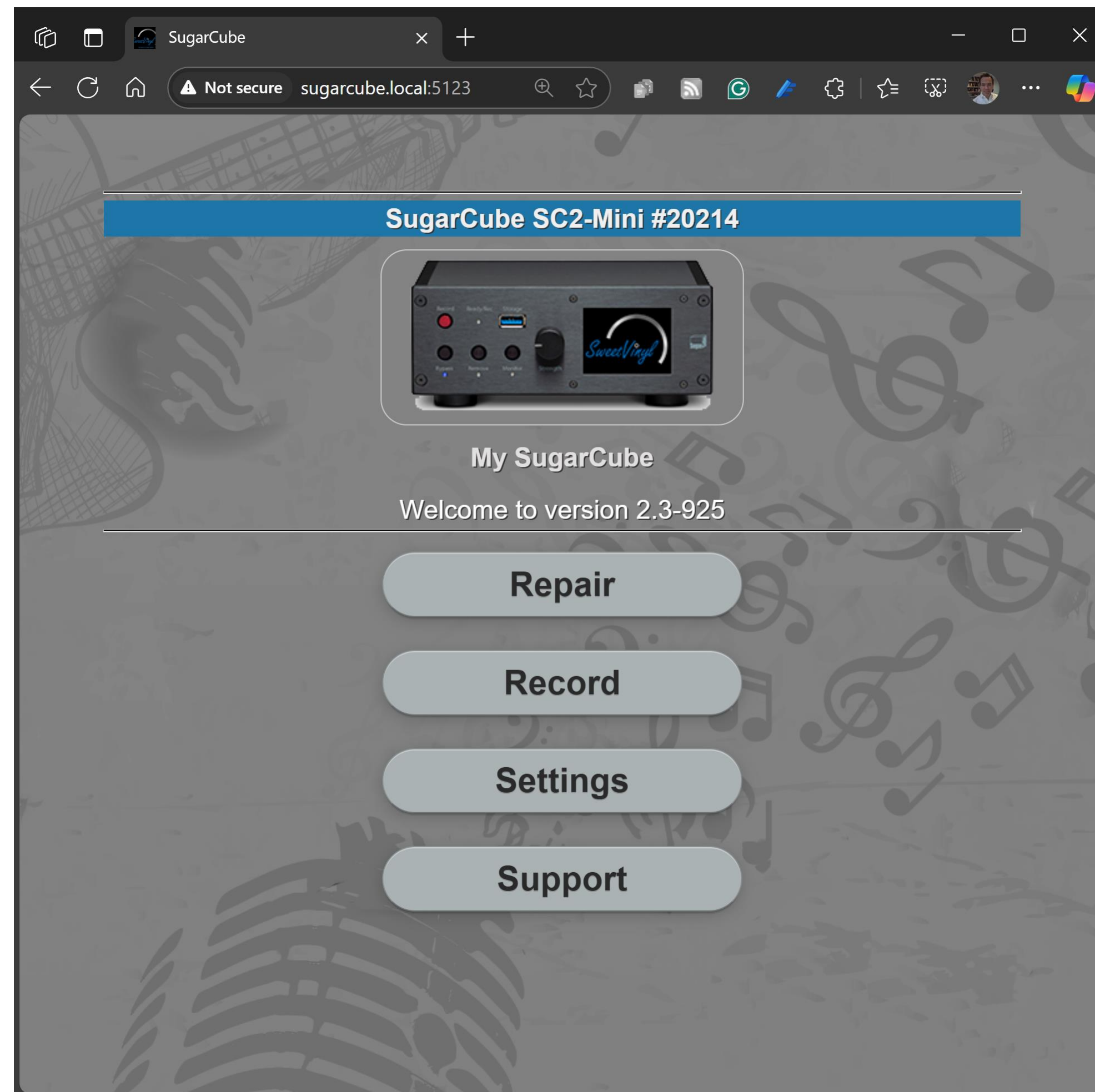
3. User Interface

MacOS, Windows
Linux, iOS, or Android



14. SugarCube Functions

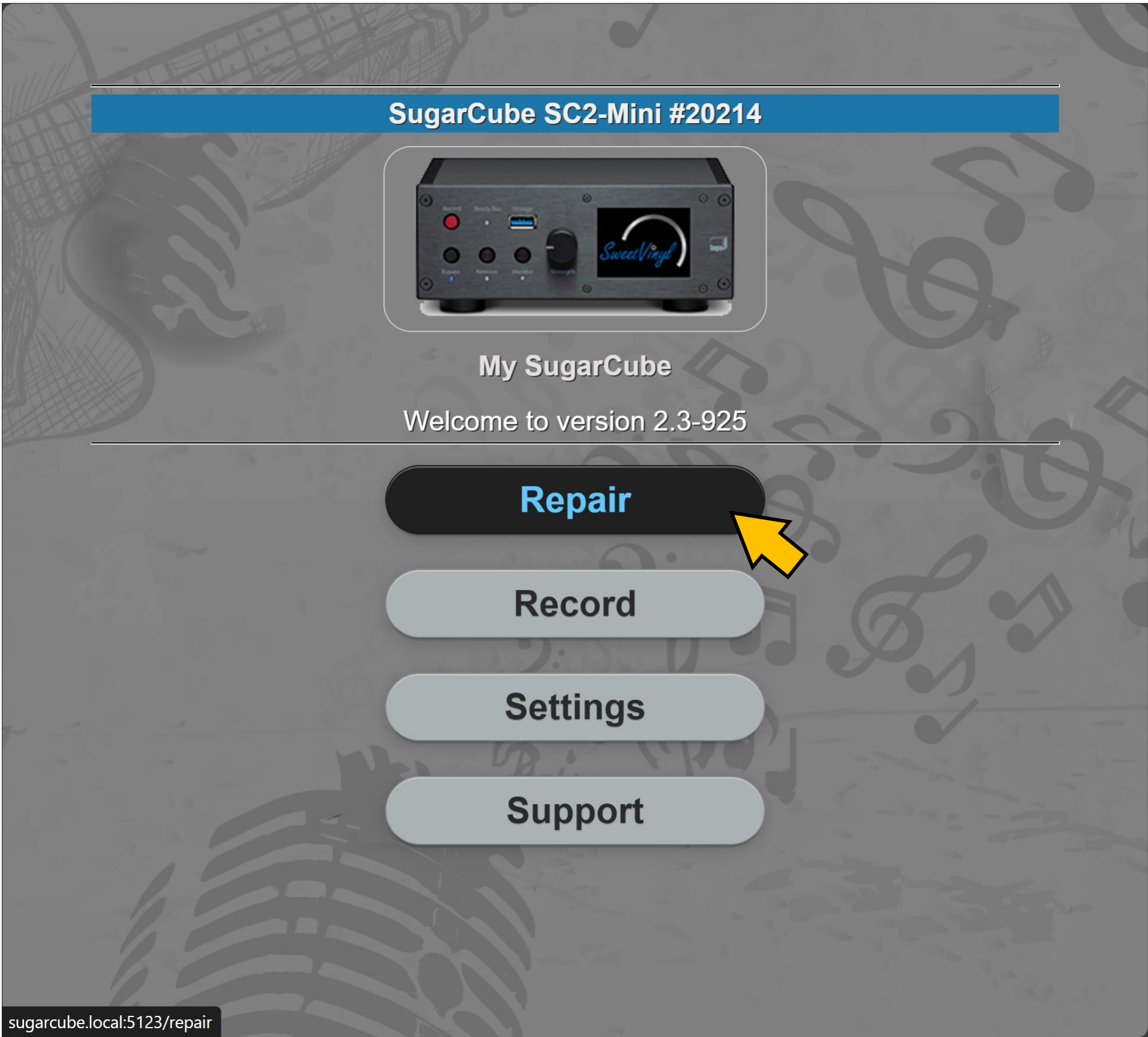
Main Page of User Interface



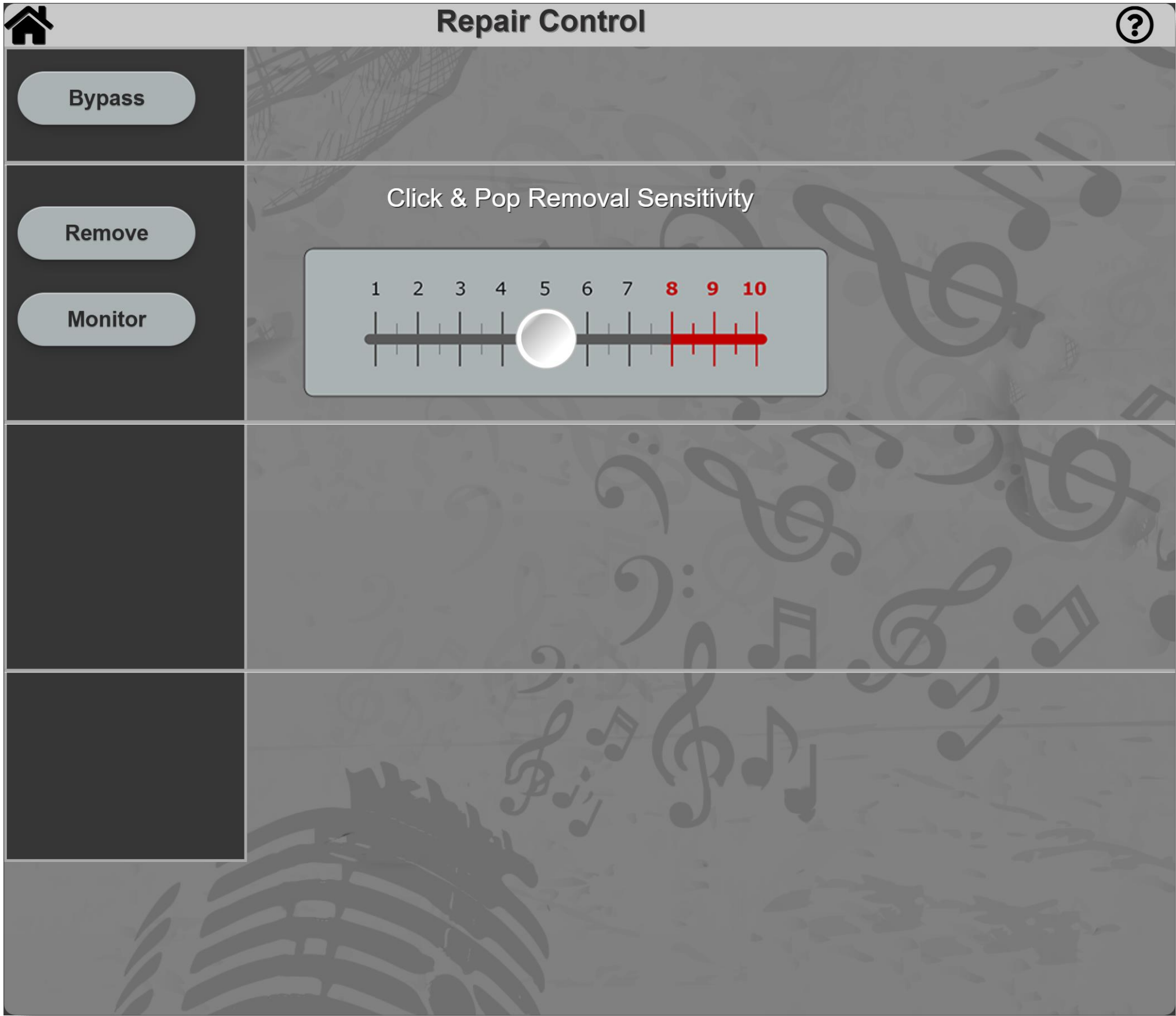
(1) This is the 'home' page from which you access all functions

18. SugarCube Functions

Repairing Clicks and Pops



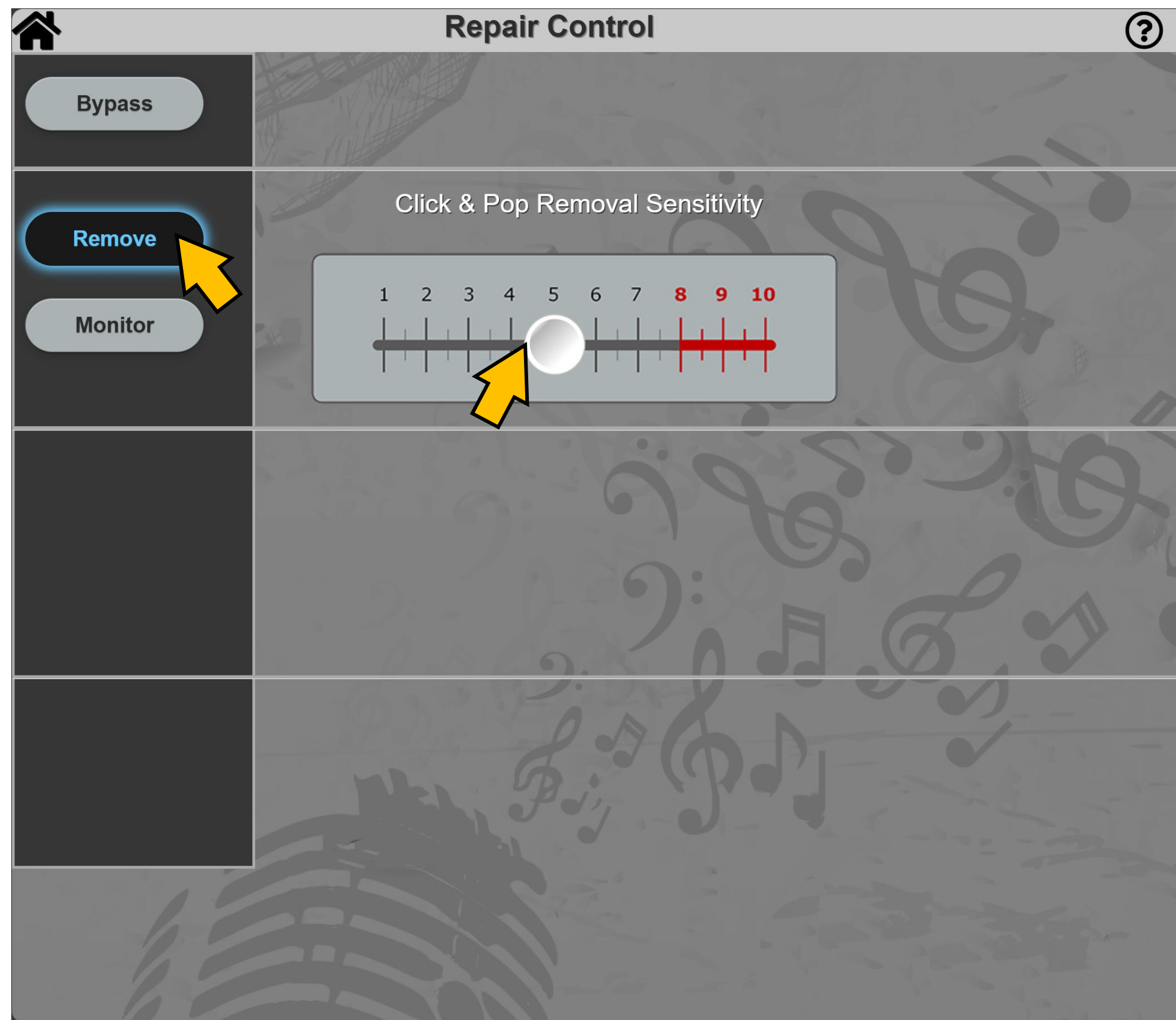
(1) Click on 'Repair'



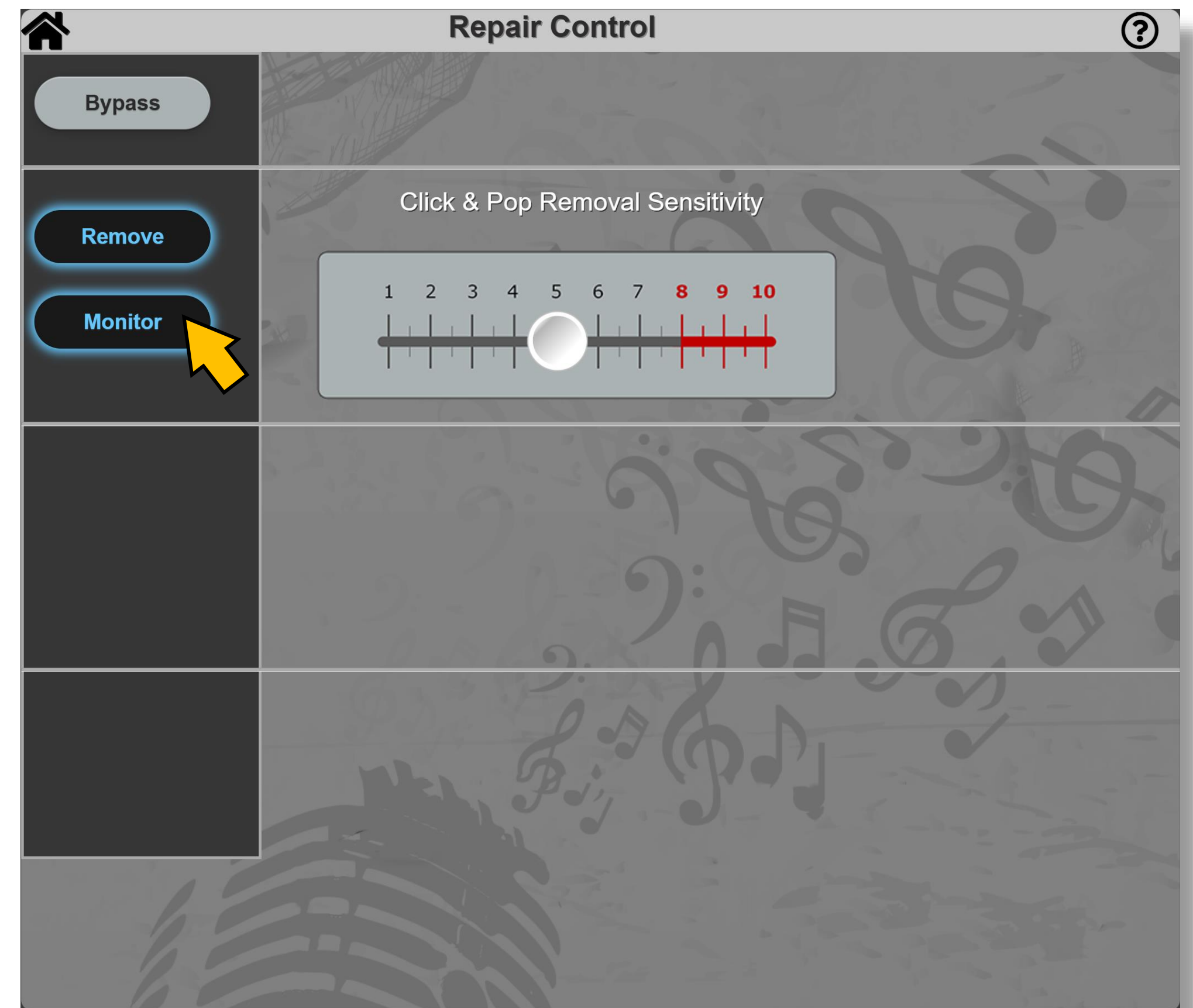
(2) These functions mirror the front panel controls. A change here is reflected on the front panel and a change on the front panel changes the above.

19. SugarCube Functions

Repairing Clicks and Pops



- (1) Click on 'Remove' to engage the click and pop removal process
- (2) Slide the control left or right to adjust sensitivity
NOTE: this control is locked to the front panel control. When you turn the dial on the front panel, the slide moved in concert with it.



- (3) Click on 'Monitor' here or on the front panel to listen with audio suppressed to hear the clicks and pops being removed. This helps you set optimal 'Sensitivity'.

20. SugarCube Functions

Setting Input Level

Input Level	Headroom
+0dB	+14dB
-6dB	+20dB
-12dB	+26dB
-18dB	+32dB

Input Level Detail

Headroom estimates were derived from the ClearAudio Test LP, catalog LPT 83063, www.clearaudio.de using tracks 1-3, 1kHz test tones mastered at 0dB (8cm/sec), played back using Ortofon 2M Bronze MM cartridge with calibrated phono stage, 47k ohm load at +40dB gain

For 'Line In' level, start with '+0dB'.
If you experience clipping, step down to a lower level.

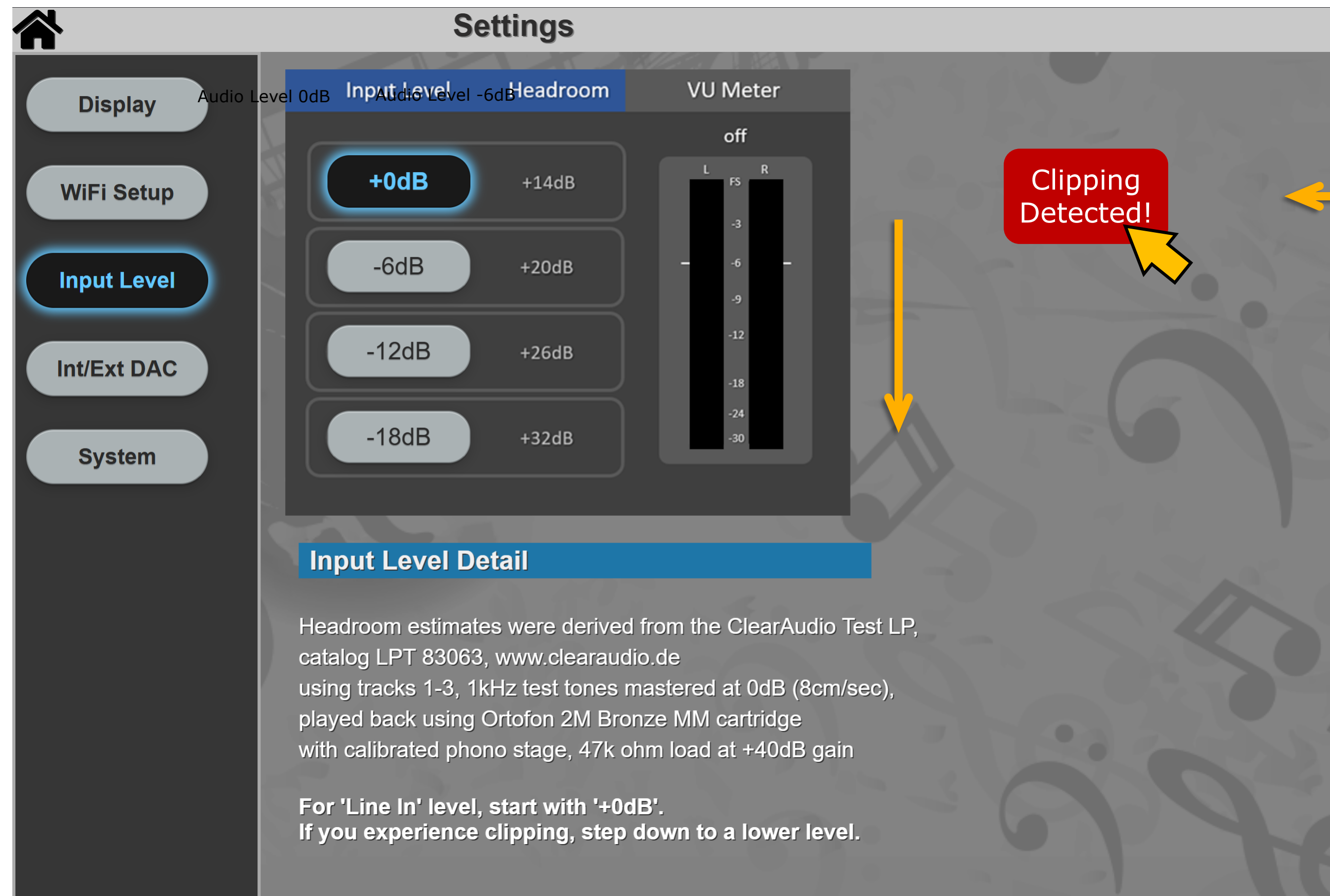
Annotations:

- Yellow arrow points to the +0dB button: This is the default operating level for almost any input. At this setting, you have +14dB of headroom for transients (e.g.-- a stick hitting a cymbal)
- Yellow arrow points to the -6dB button: If the phono pre-amp has a high output, step down -6dB to avoid clipping or if you want even more headroom with a normal input.
- Yellow arrow points to the -12dB button: If set to the above and the 'Clipping' alert is seen, select this.
- Yellow arrow points to the -18dB button: If set to the above and the 'Clipping' alert is seen, select this.

IMPORTANT NOTE: The output of the SugarCube will always equal the input. Changes in settings here will result in NO change in volume of **Line Out** of the SugarCube.

21. SugarCube Functions

Managing a 'Clipping' Warning

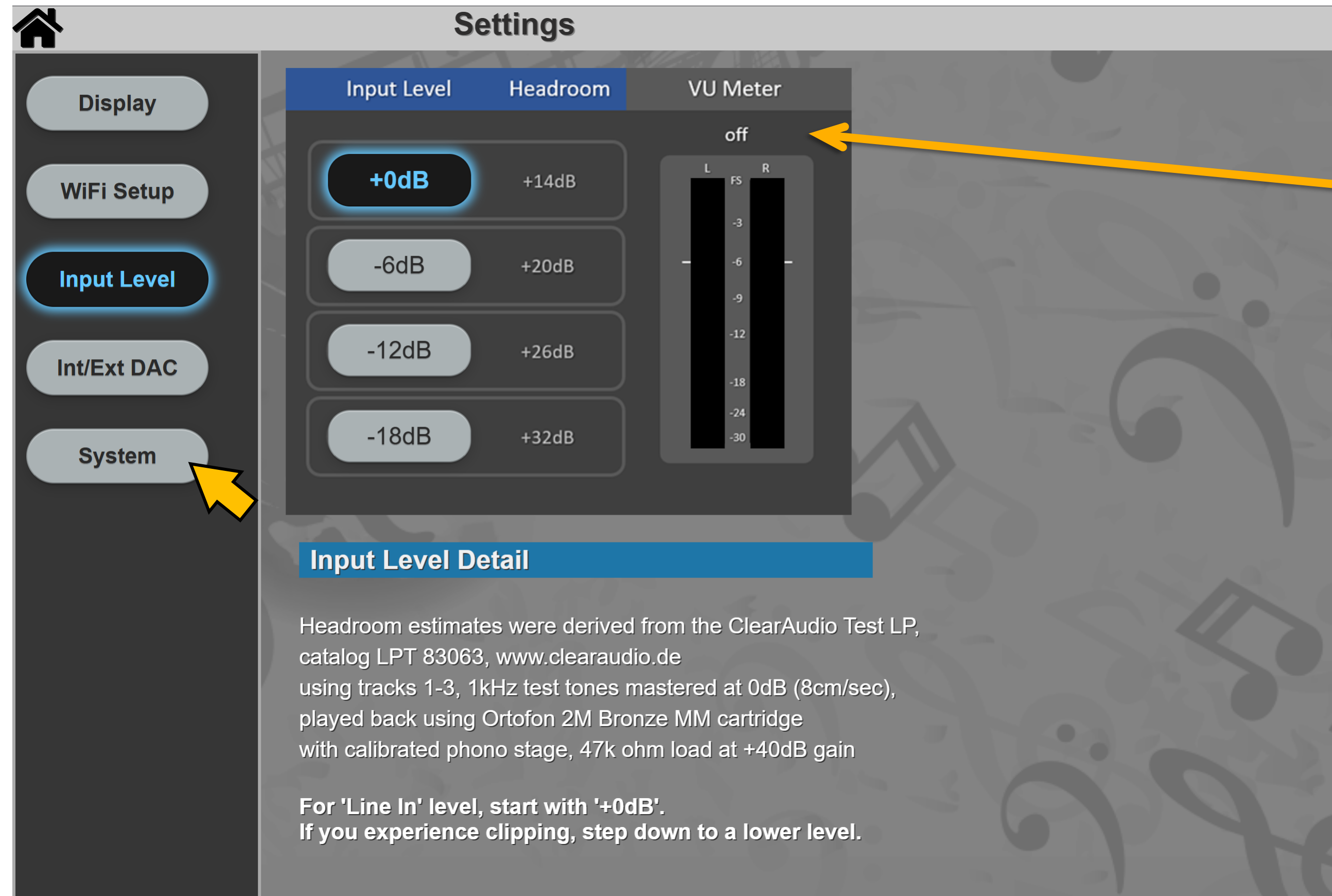


NOTE: if this appears when, for example, you just dropped the needle on an LP, just click on it and it will go away.

However: if you see this warning while an LP is playing, then move down a level.

21. SugarCube Functions

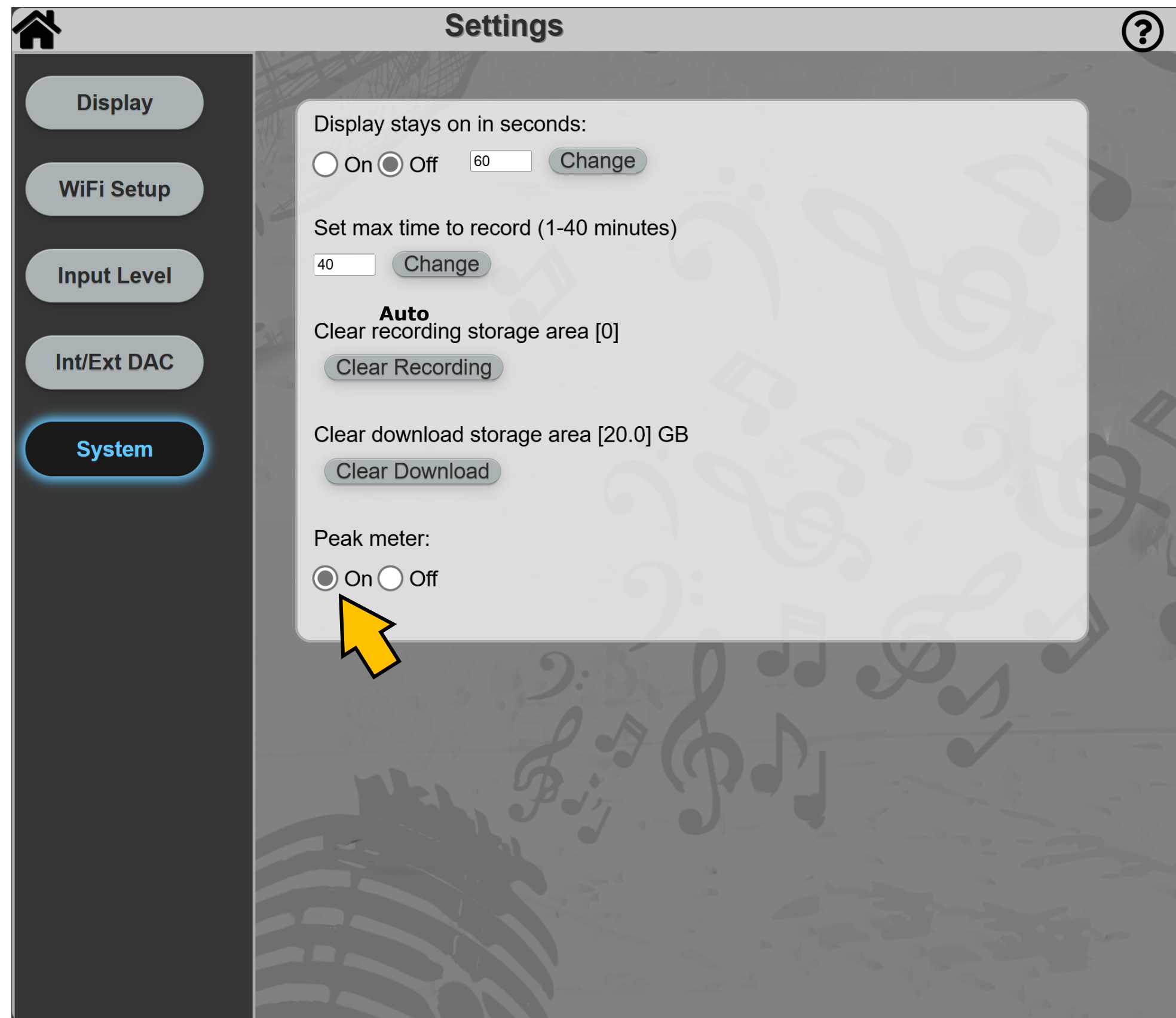
Using the VU (Peak) Meter



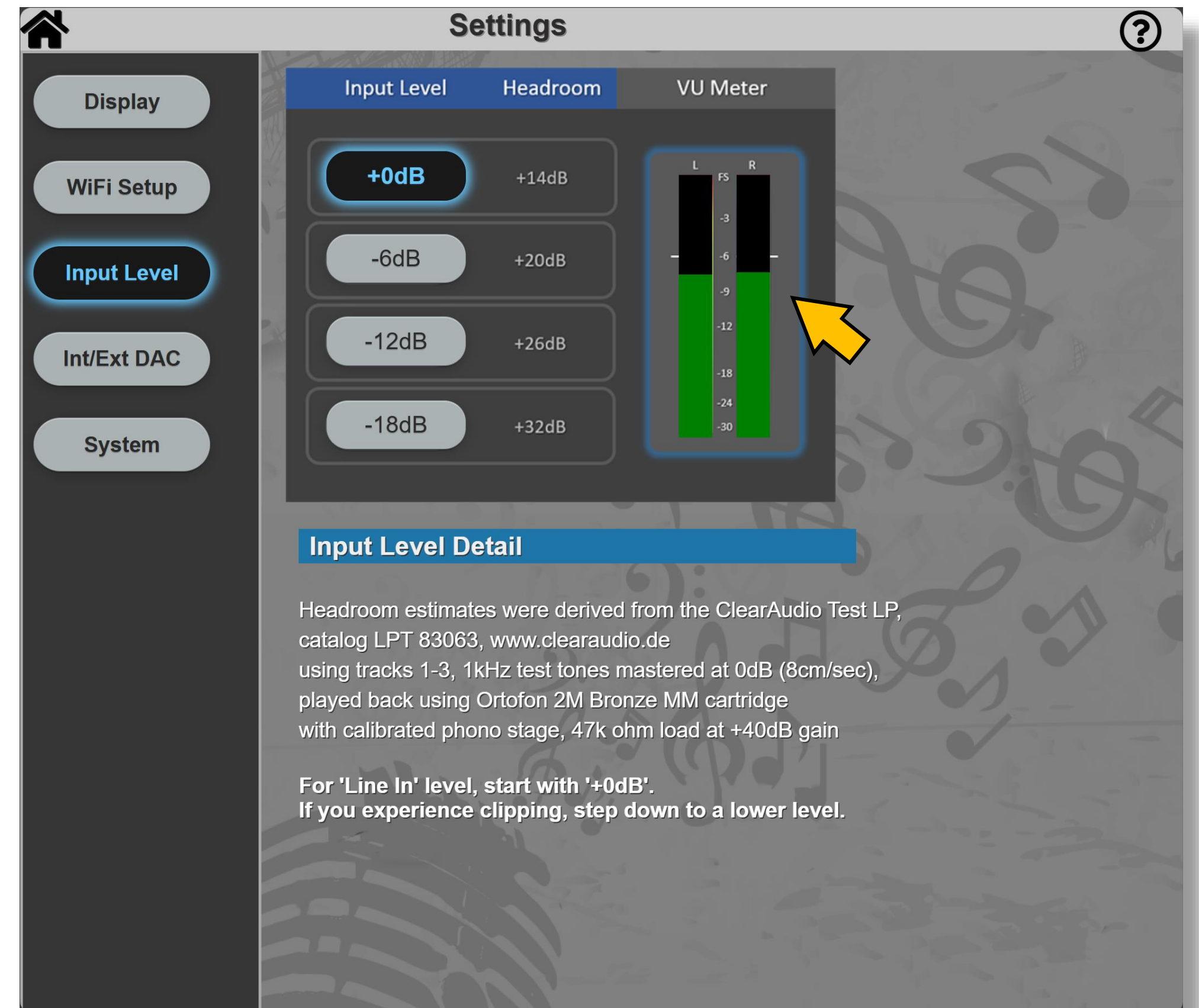
If you would like to see the live audio input, select '**System**' and see next page.

23. SugarCube Functions

Using the VU (Peak) Meter



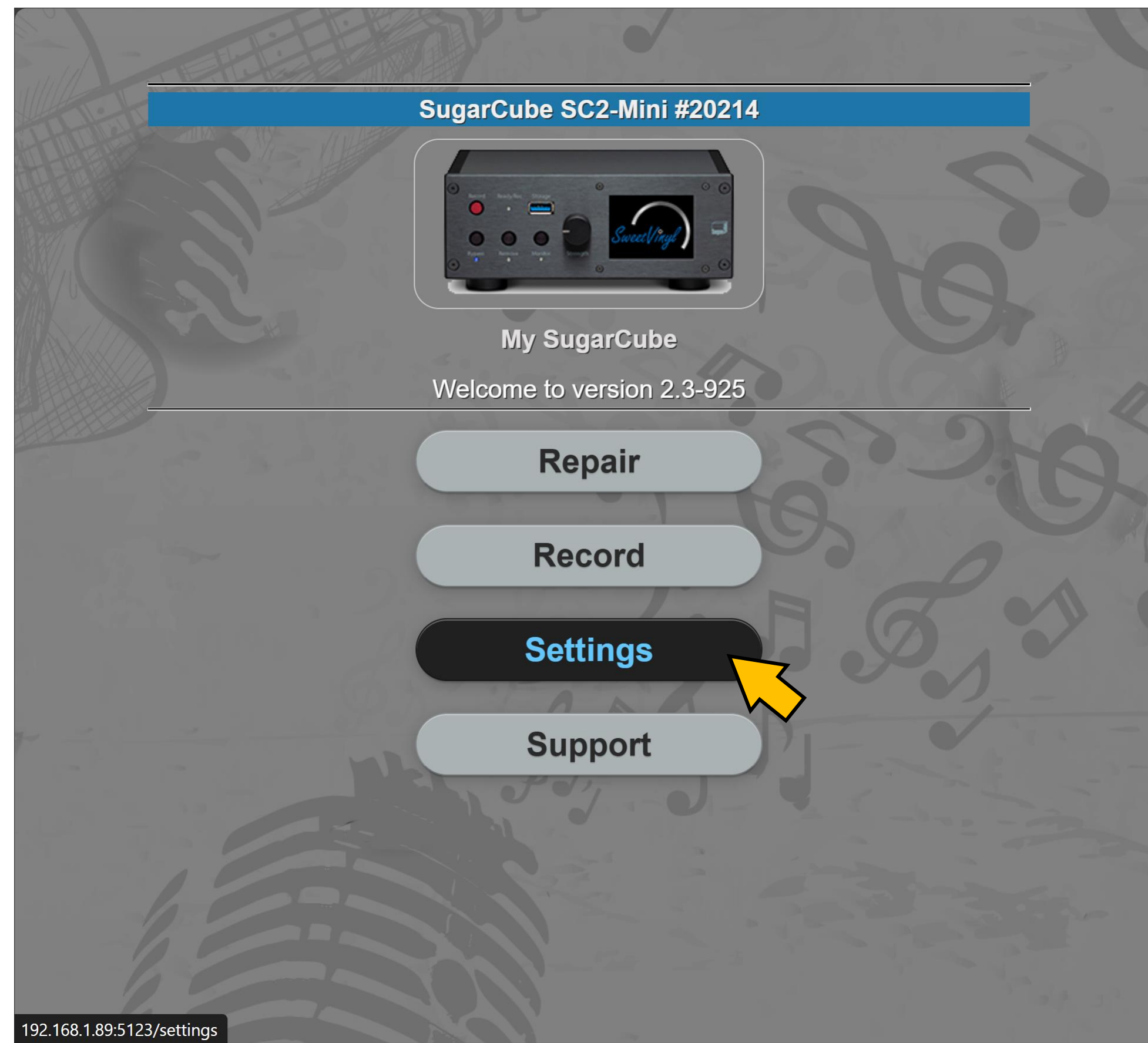
(1) This selection turns on and off the VU (Peak) Meter. ***We recommend that you turn it off after using it*** (you won't see it during operations, anyway.)



(2) Audio levels display here (currently showing equal test tones on inputs.)

22. SugarCube Functions

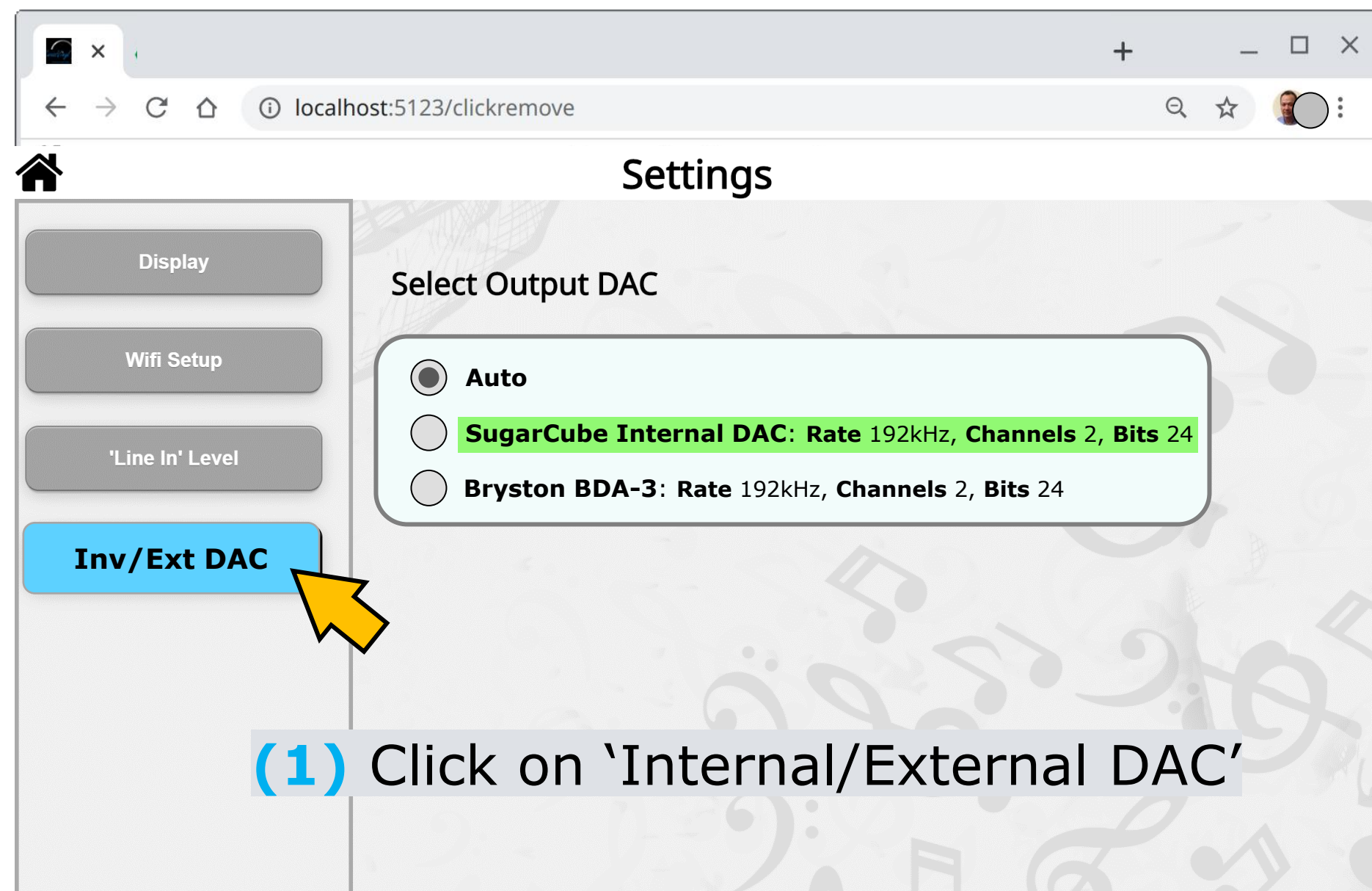
Selecting an External DAC



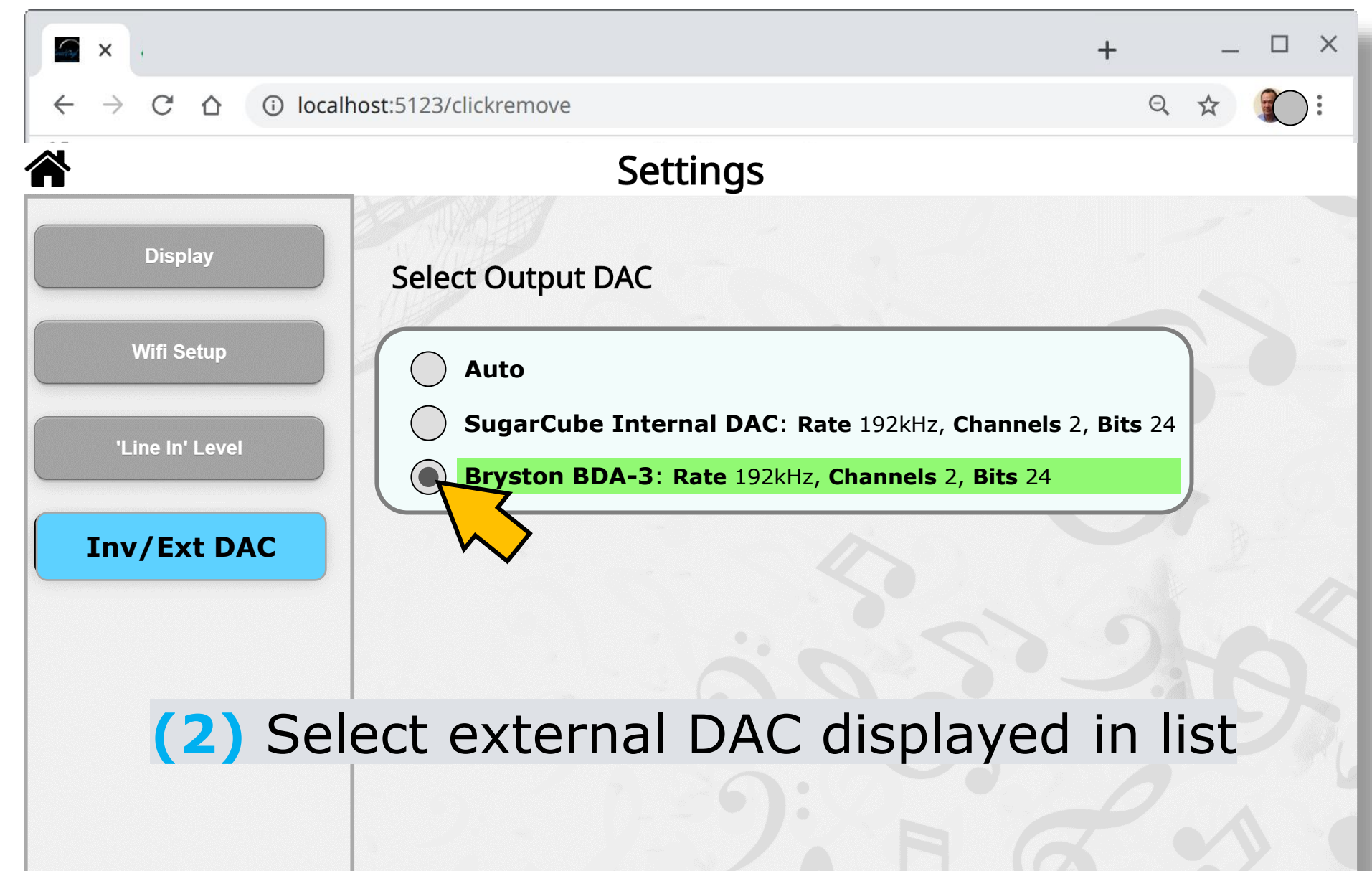
(1) Click on 'Settings'

23. SugarCube Functions

Selecting an External DAC



(1) Click on 'Internal/External DAC'



(2) Select external DAC displayed in list

Please read before using an external DAC: The USB output on the back of the SugarCube supports '**USB Class 2 Audio**' operation which is today almost universally used for consumer applications. However, USB Class 2 Audio is an **asynchronous protocol**, meaning it carries **NO CLOCK**.

For ordinary digital playback of a digital audio file, this does not matter because the DAC is pulling audio data from a server (which patiently waits until the next packet request from the DAC.) However, when connected to the SugarCube, the DAC no longer has control and must take audio packets that are being pushed to it by the SugarCube's ADC.

The internal crystal clock of the SugarCube provides the clock reference for both its internal ADC and the DAC so the two are always in perfect sync. However, as the USB Class 2 Audio output carries no clock (by definition), the internal clock of your DAC becomes its clock reference for playback.

Typical crystal clock references are accurate within $\pm 100\text{ppm}$ of its target frequency which means that worst case at 192kHz sample rate, every 0.32 seconds the SugarCube and your DAC will be off by one sample. Depending on how well your DAC handles dropped data, there is a possibility of an audible click when your DAC gives up and does a resync, anywhere between 5 to 15 minutes. Bottom line: use an external DAC at your discretion.

Setting Up SugarCube Mobile App

4. Connecting to the SugarCube

Apple iOS and Android
Guide





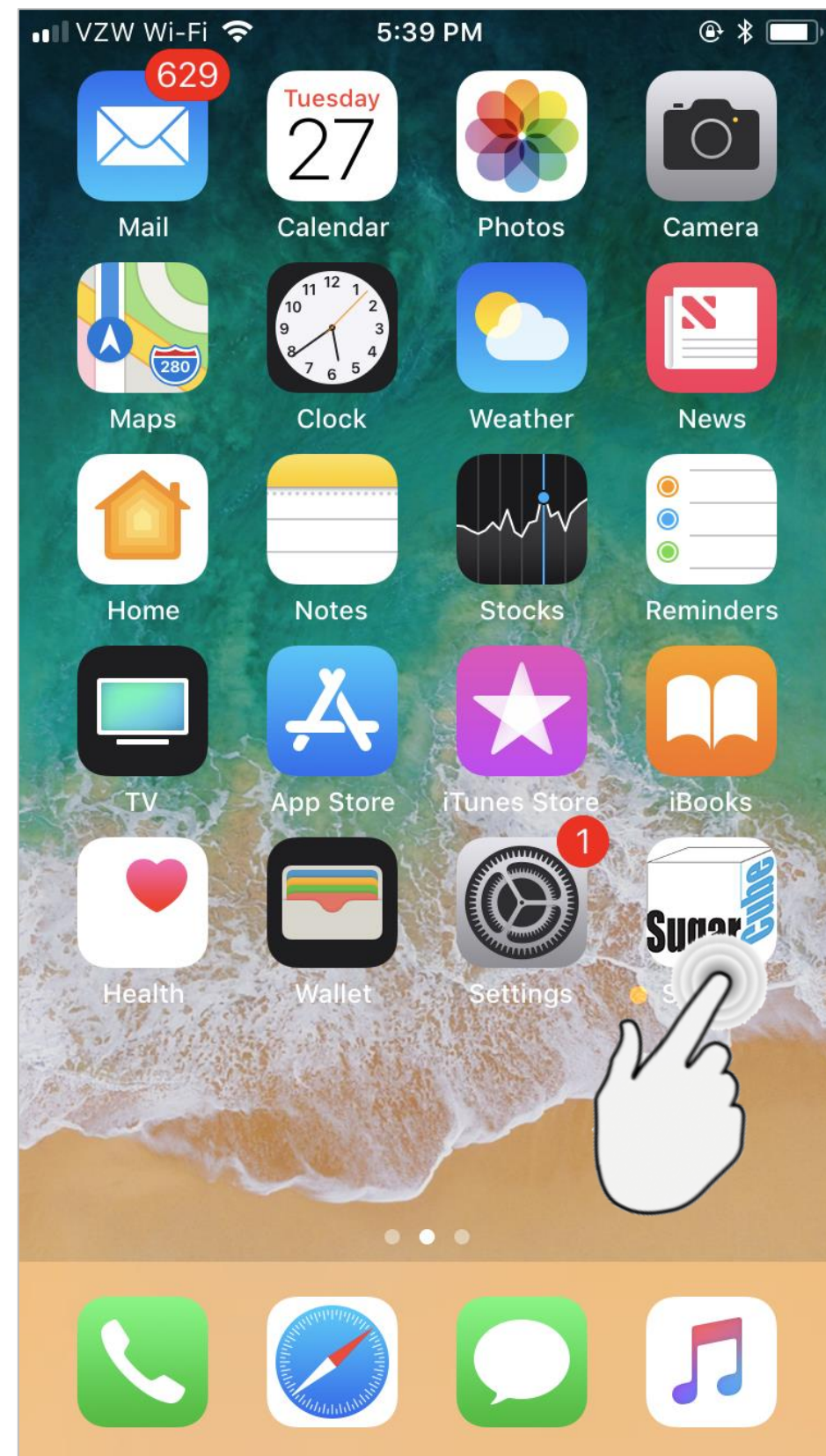
Using an Apple Mobile Device (Apple iOS)

25. Items needed to set up Apple to control SugarCube:

1. iPhone or iPad.
2. A network with WiFi your Apple device can communicate with and that can be connected to your SugarCube with WiFi or wired Ethernet.
3. From the Apple App Store, you will be able to download the 'SugarCube' app to your Apple mobile device. (Look for: SugarCube by SweetVinyl.)
4. The WiFi dongle supplied with your SugarCube.

Note: If you plan to use **wired Ethernet**, there is no need to plug in the WiFi adapter to the SugarCube; please skip to **Section 1.3** and follow the instructions there.

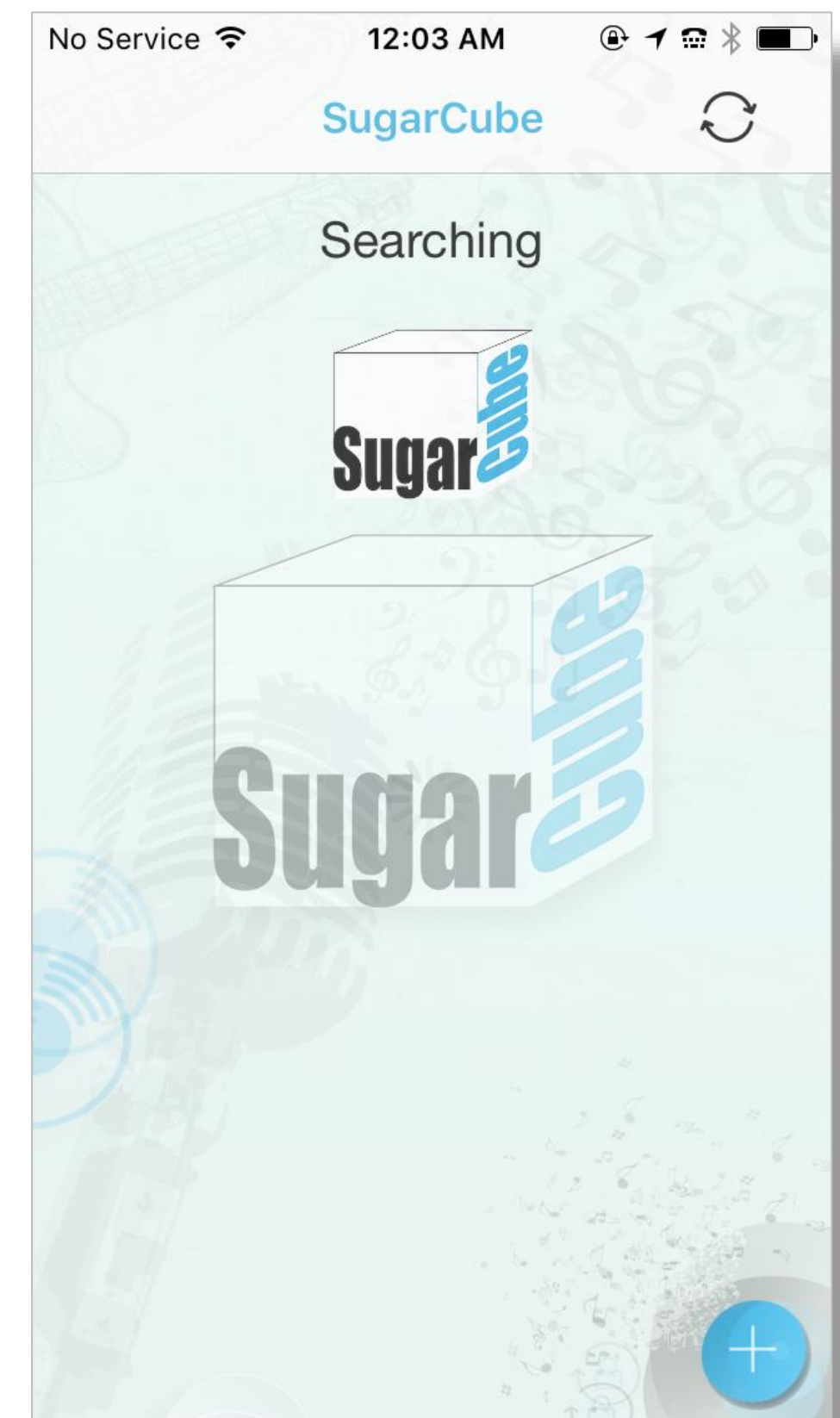
26. Connecting the app with SugarCube on Wi-Fi



(1) Launch the SugarCube app

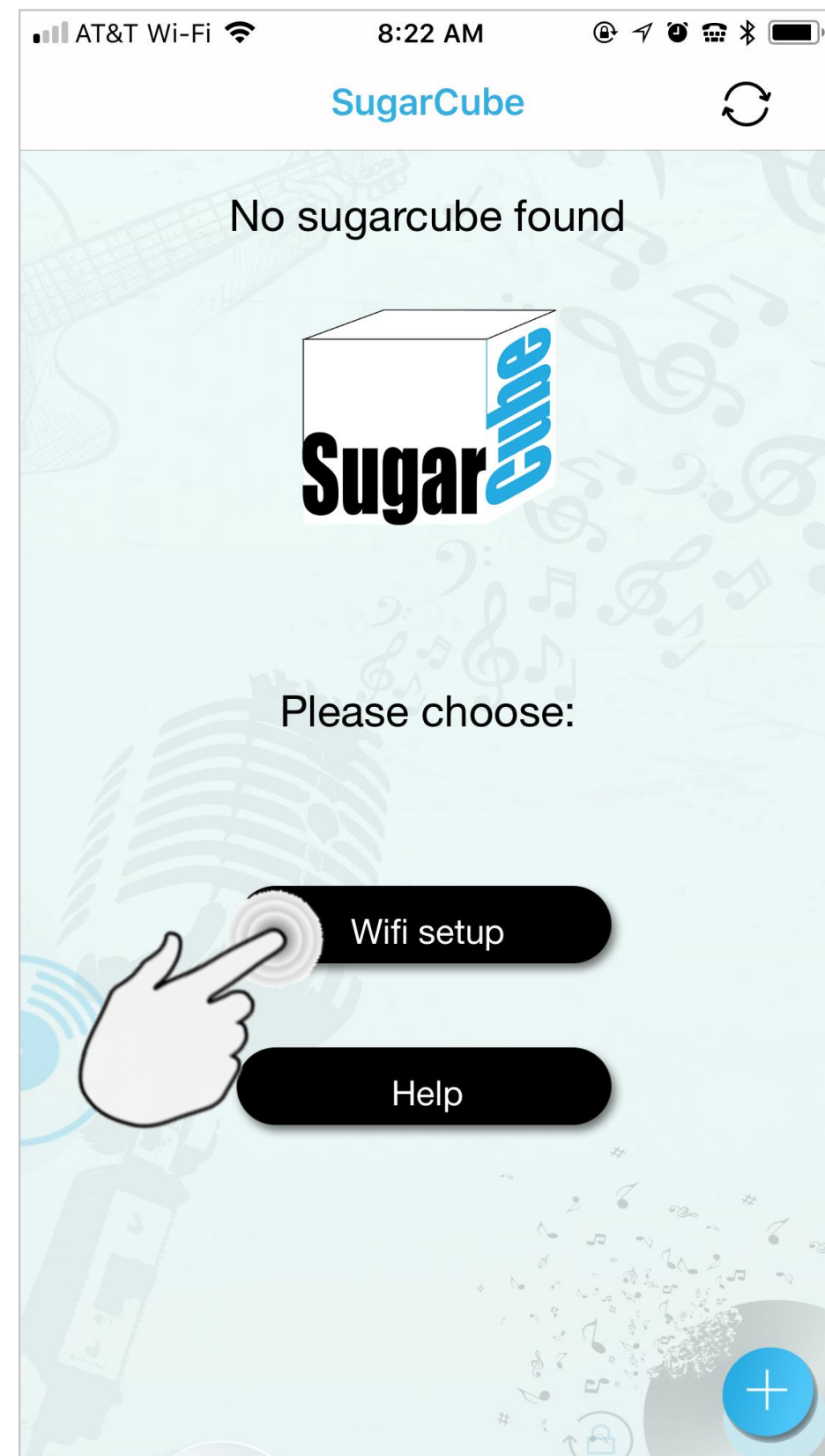


(2) App is loading

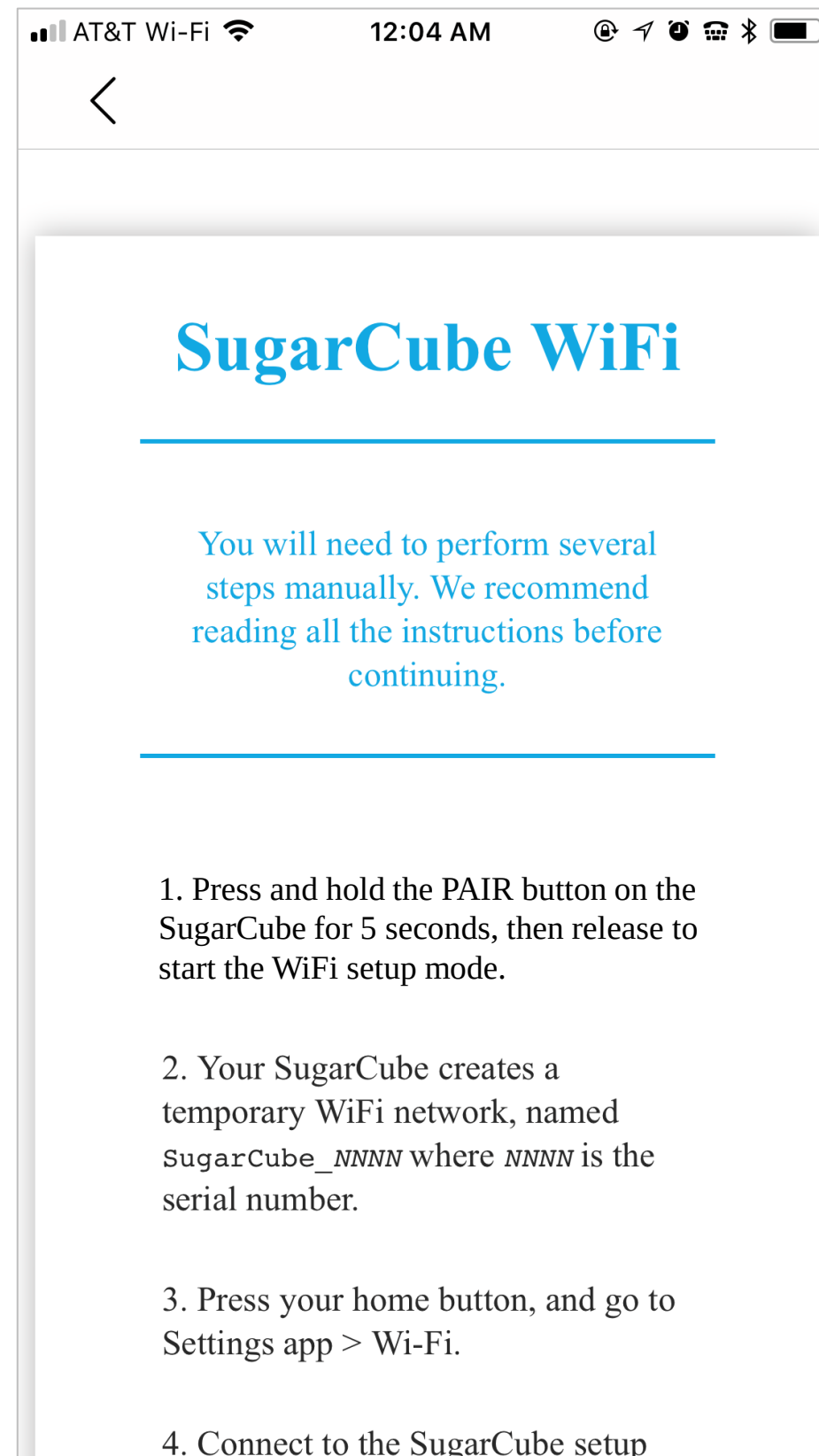


(3) App is searching for SugarCubes already setup on the LAN if any. For the first time user, nothing will be found, of course.

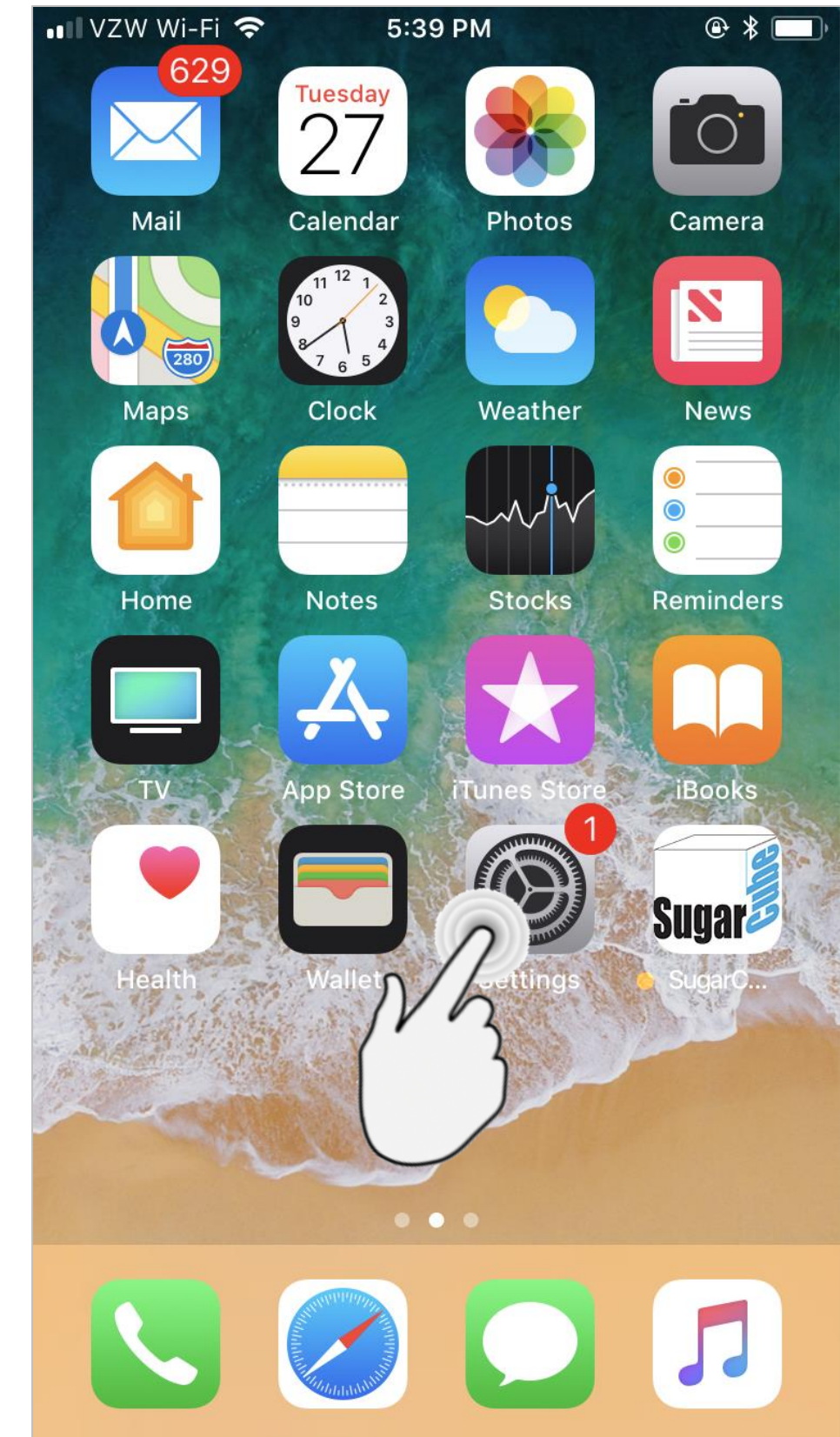
27. Connecting the app with SugarCube on Wi-Fi



(4) No SugarCube is found so the app offers the option to run WiFi setup.

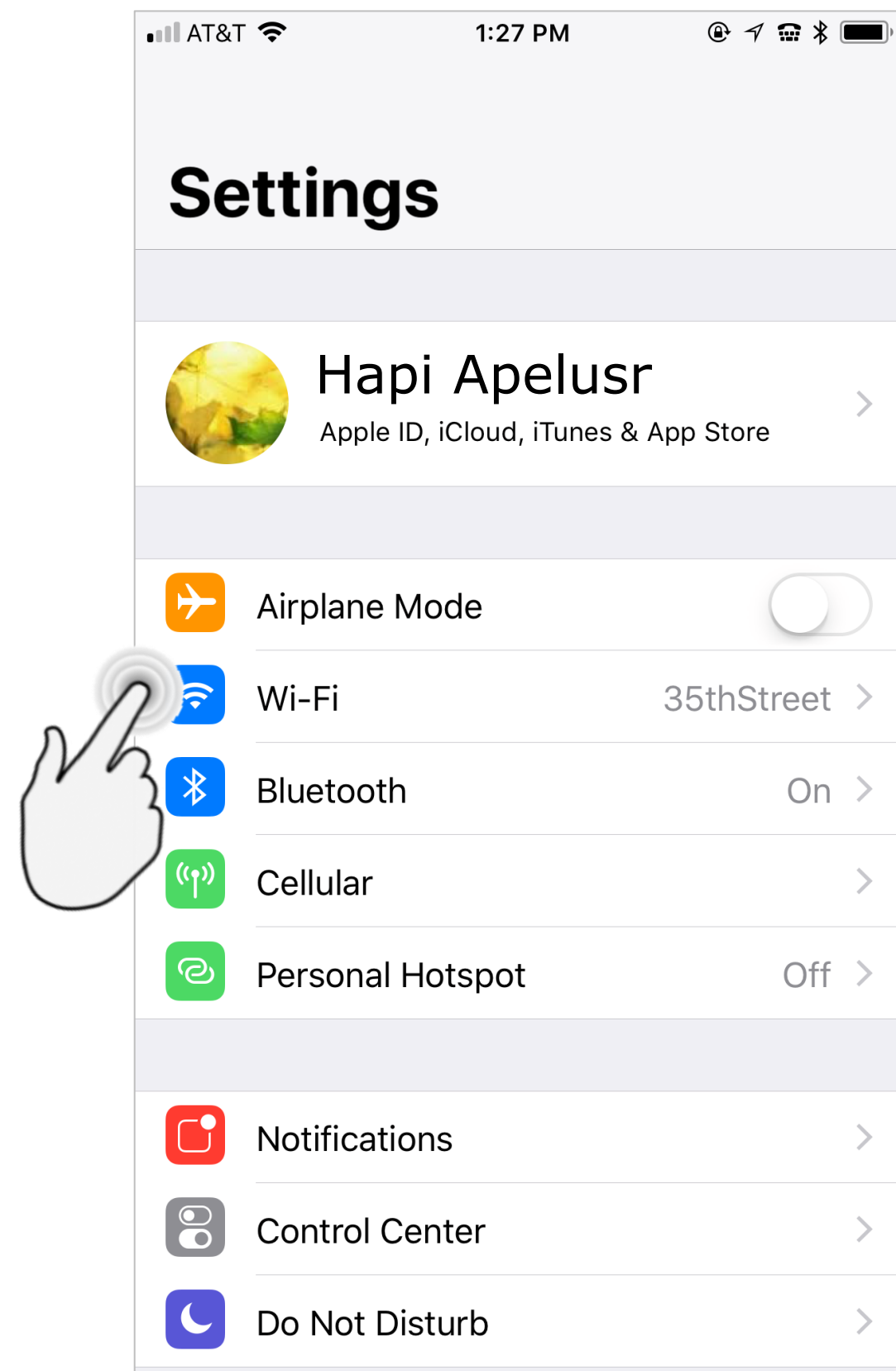


(5) The user is instructed to press and hold the front panel knob for 5 seconds to put the SugarCube into 'WiFi Setup Mode' which will be displayed on the front panel.

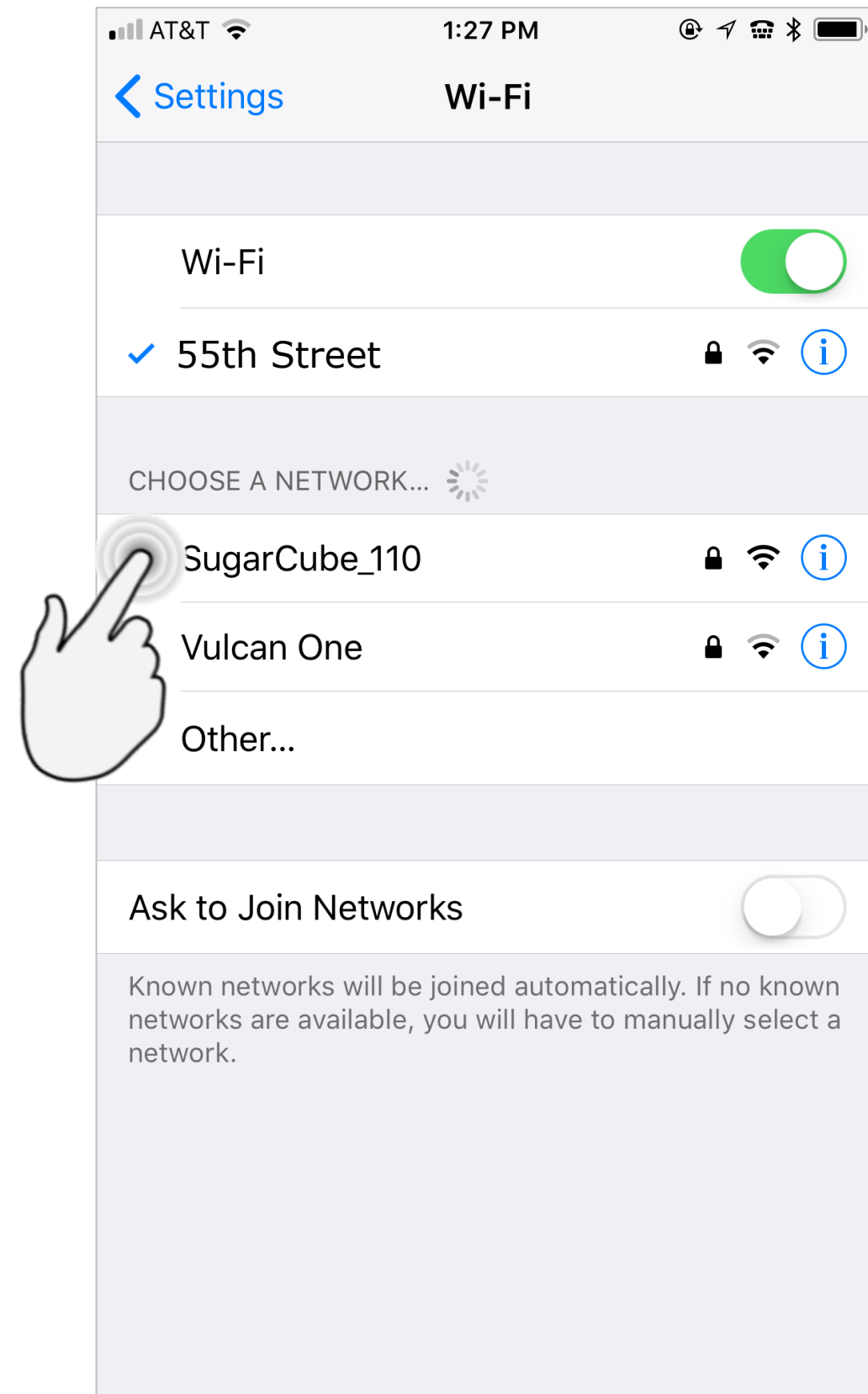


(6) The instructions direct the user to press the 'Home' key to get back to the 'Home' screen and then launch 'Setup'.

28. Connecting the app with SugarCube on WiFi

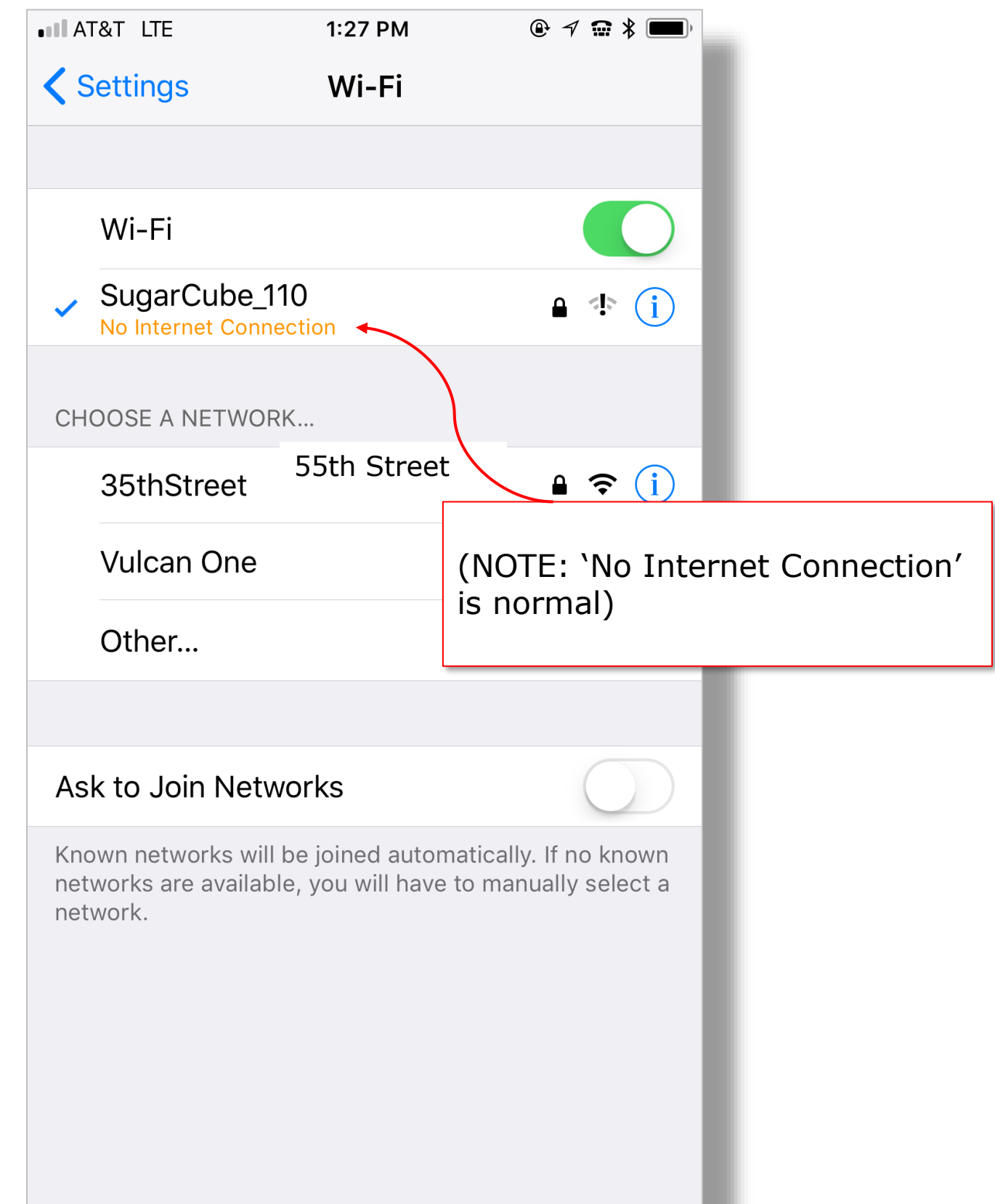


(7) When in 'Settings', select 'Wi-Fi'



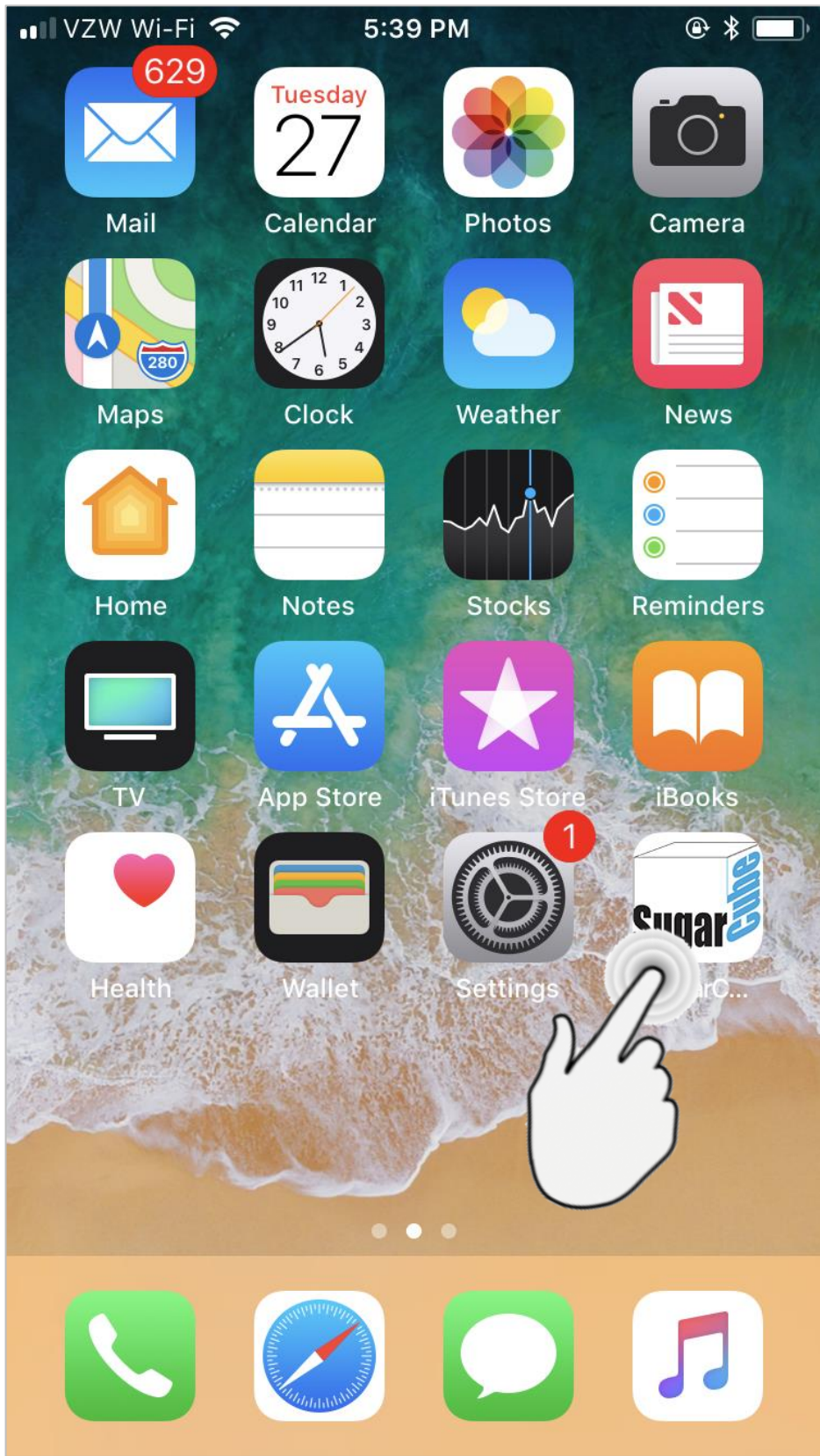
(8) Select 'SugarCube_nnnn', (whatever number appears after 'SugarCube_' is your device's serial number)

The password is: **SugarCube**

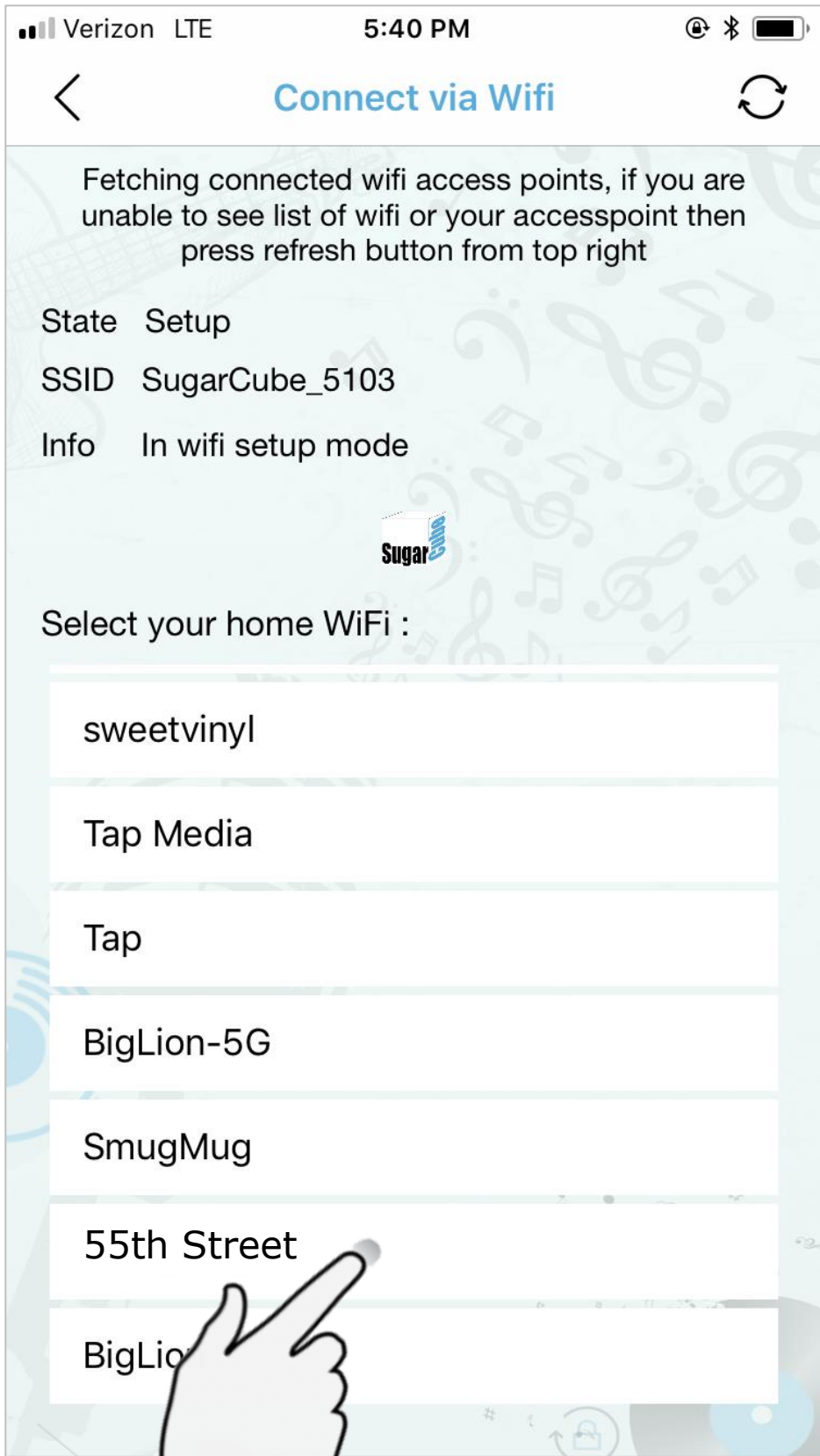


(9) With the 'SugarCube_nnnn' selected, your SugarCube is now temporarily acting as an access point.

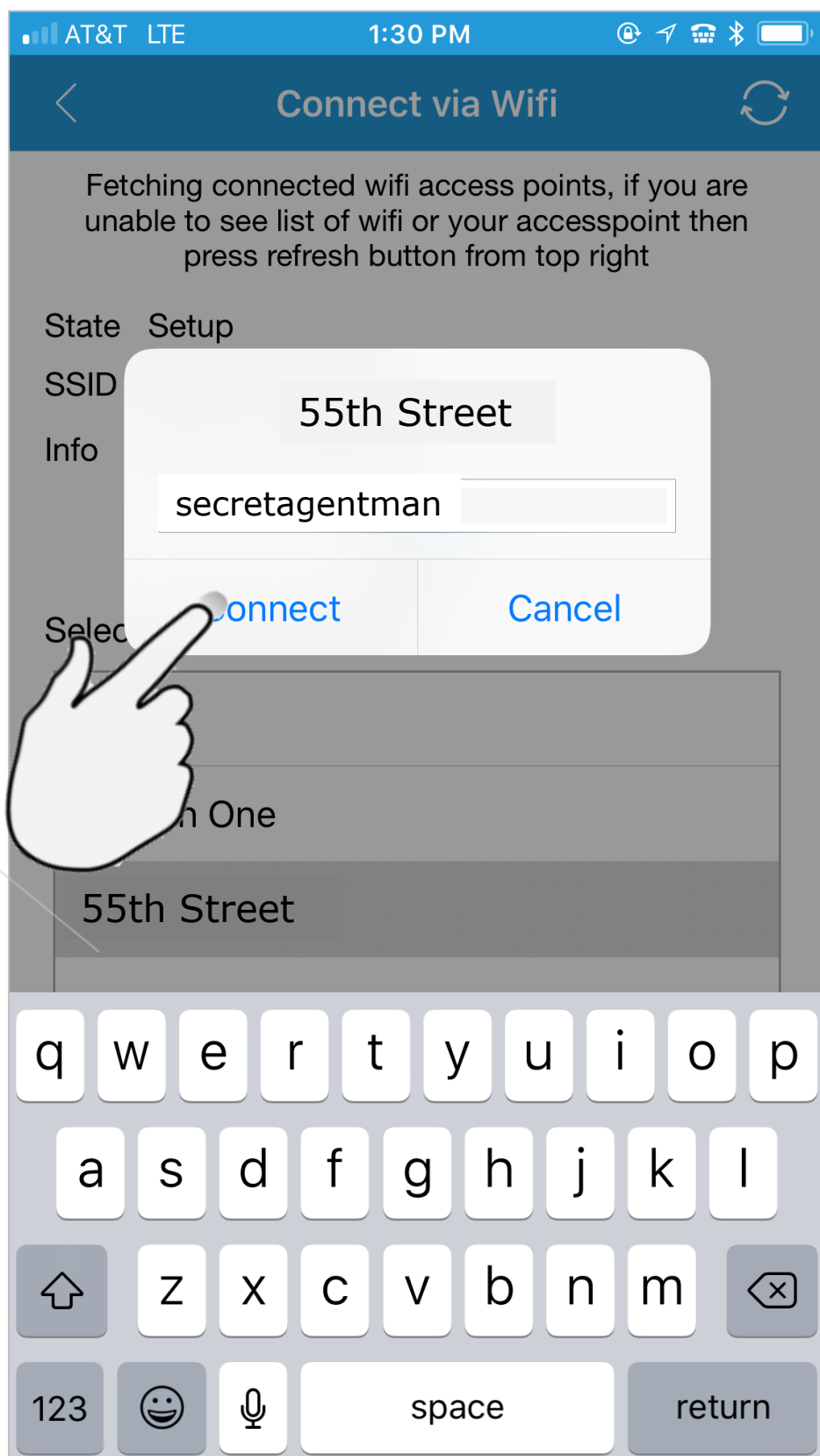
29. Connecting the app with SugarCube on Wi-Fi



(10) Relaunch the SugarCube app

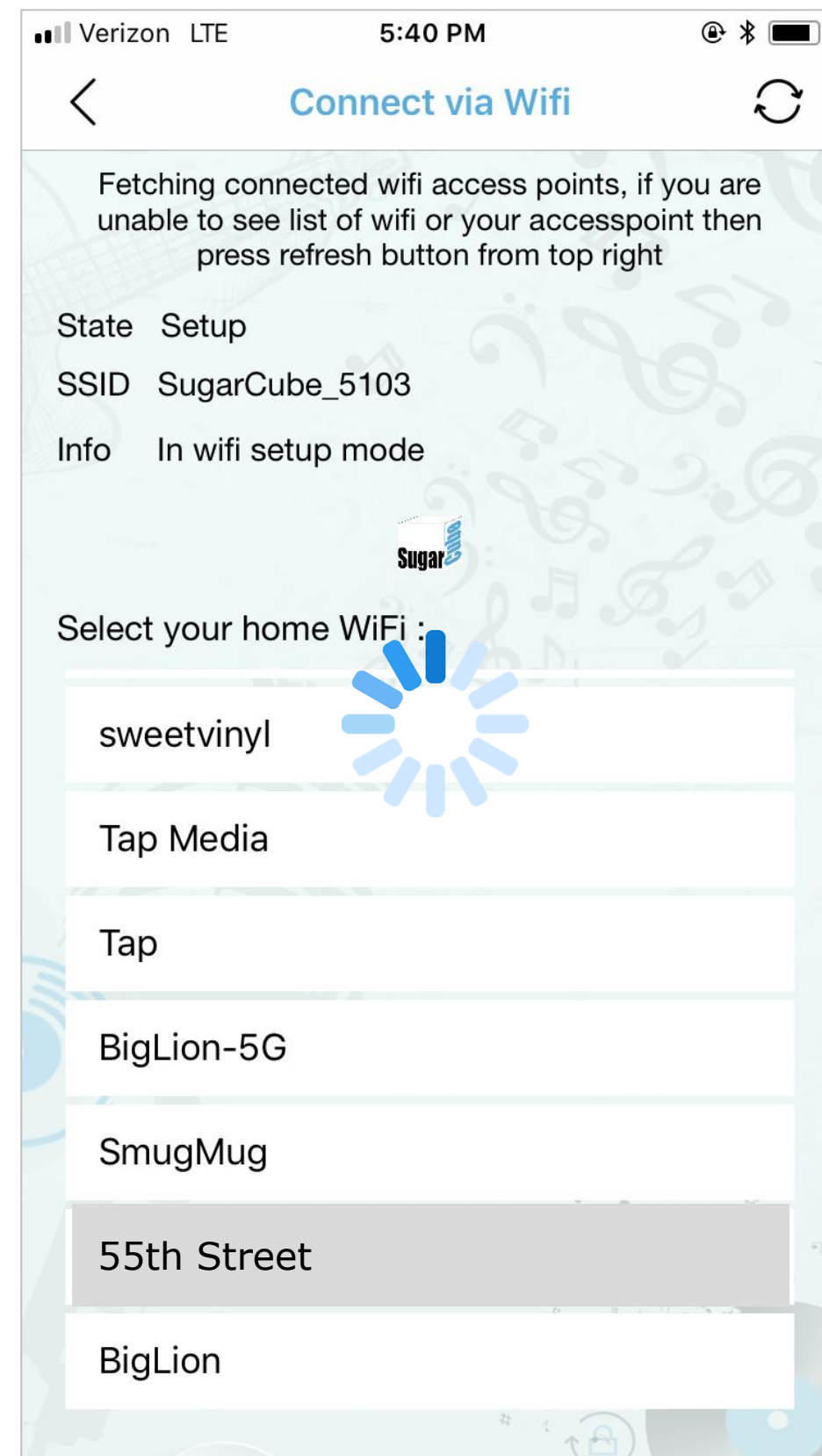


(11) Select the WiFi network that you wish to use



(12) Enter the WiFi password for your network

30. Connecting the app with SugarCube on Wi-Fi



(13) The app will now complete the connection to your WiFi network and also do a little housekeeping.

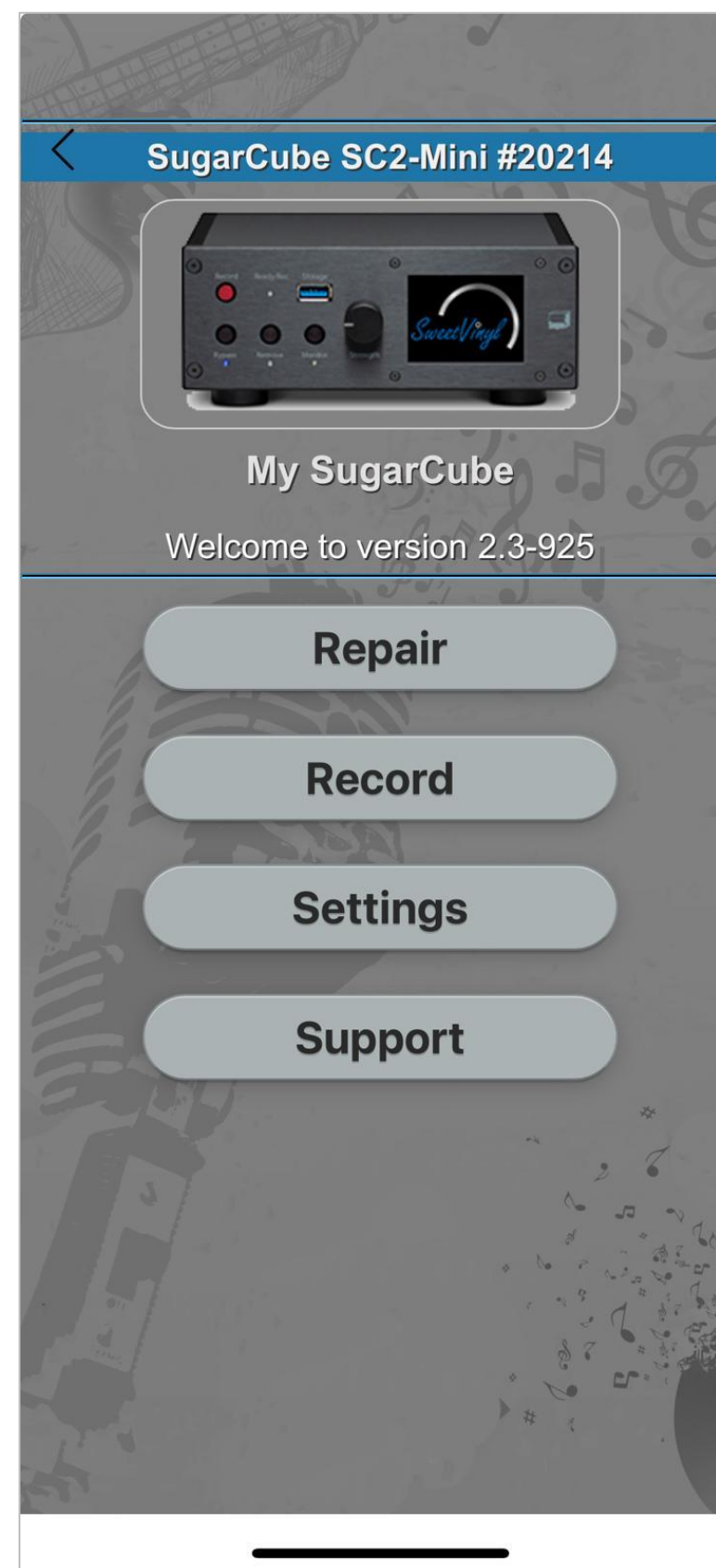


(14) The app will then display the SugarCube(s) found on your network. Select the appropriate device.



(15) If you have more than one SugarCube, press the front panel 'Sensitivity' knob to get a pairing code. Otherwise, just enter '1111'.

31. Connecting the app with SugarCube on Wi-Fi



(16) You will then see the 'Home' screen and you are done, congratulations!



Using an **Android** Mobile Device

32. Items needed to set up Android device to control SugarCube:

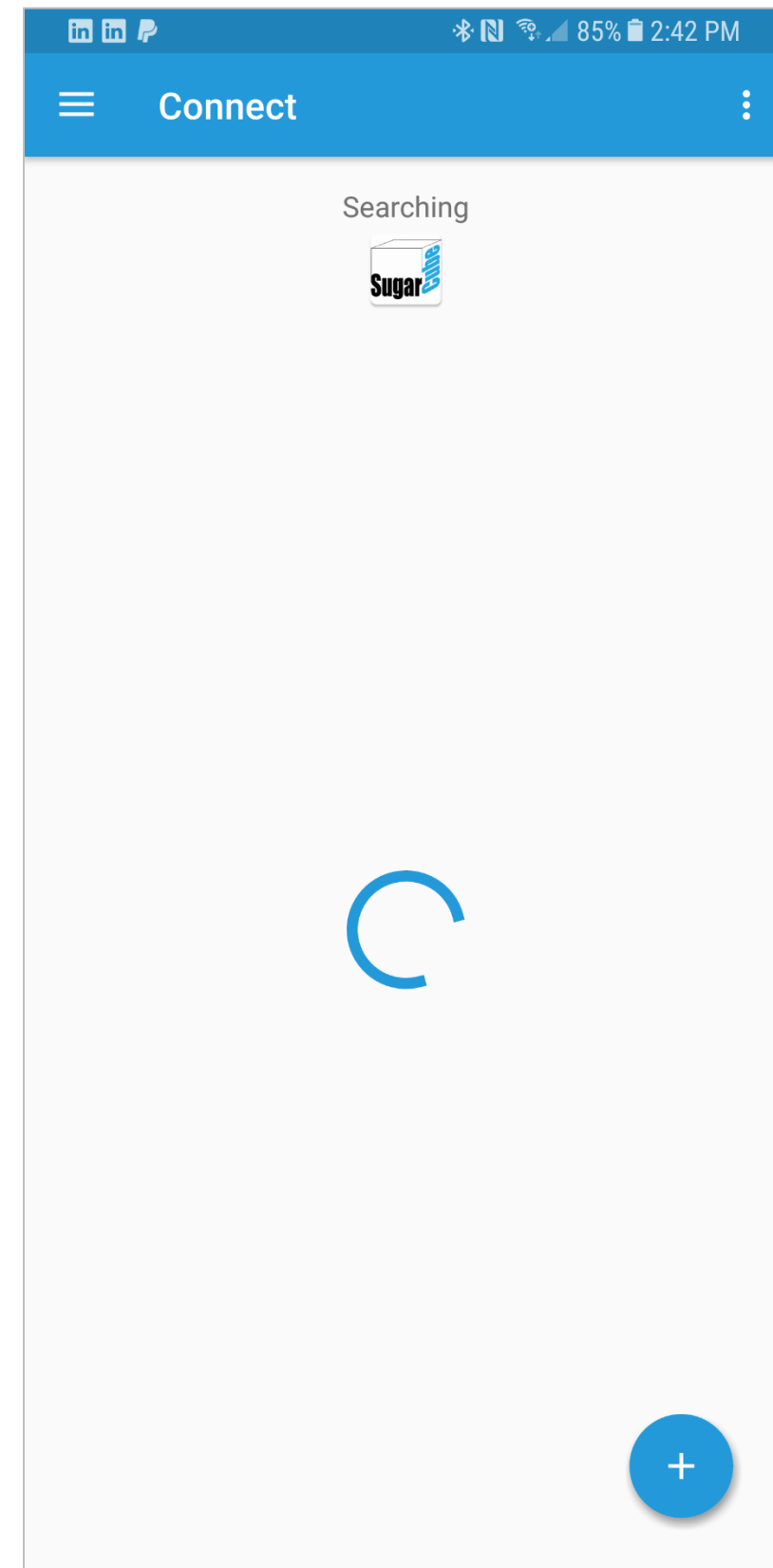
1. Android phone or tablet.
2. Network with WiFi your Android device can communicate with and that can be connected to your SugarCube with WiFi or wired Ethernet.
3. Download the Android App from the Google Play Store.
(Look for: SugarCube by SweetVinyl.)
4. The WiFi dongle supplied with your SugarCube or a wired Ethernet cable from a port on your router.

Note: If you plan to use wired Ethernet, there is no need to plug in the WiFi adapter to the SugarCube; please skip to page 1.10 and follow the instructions.

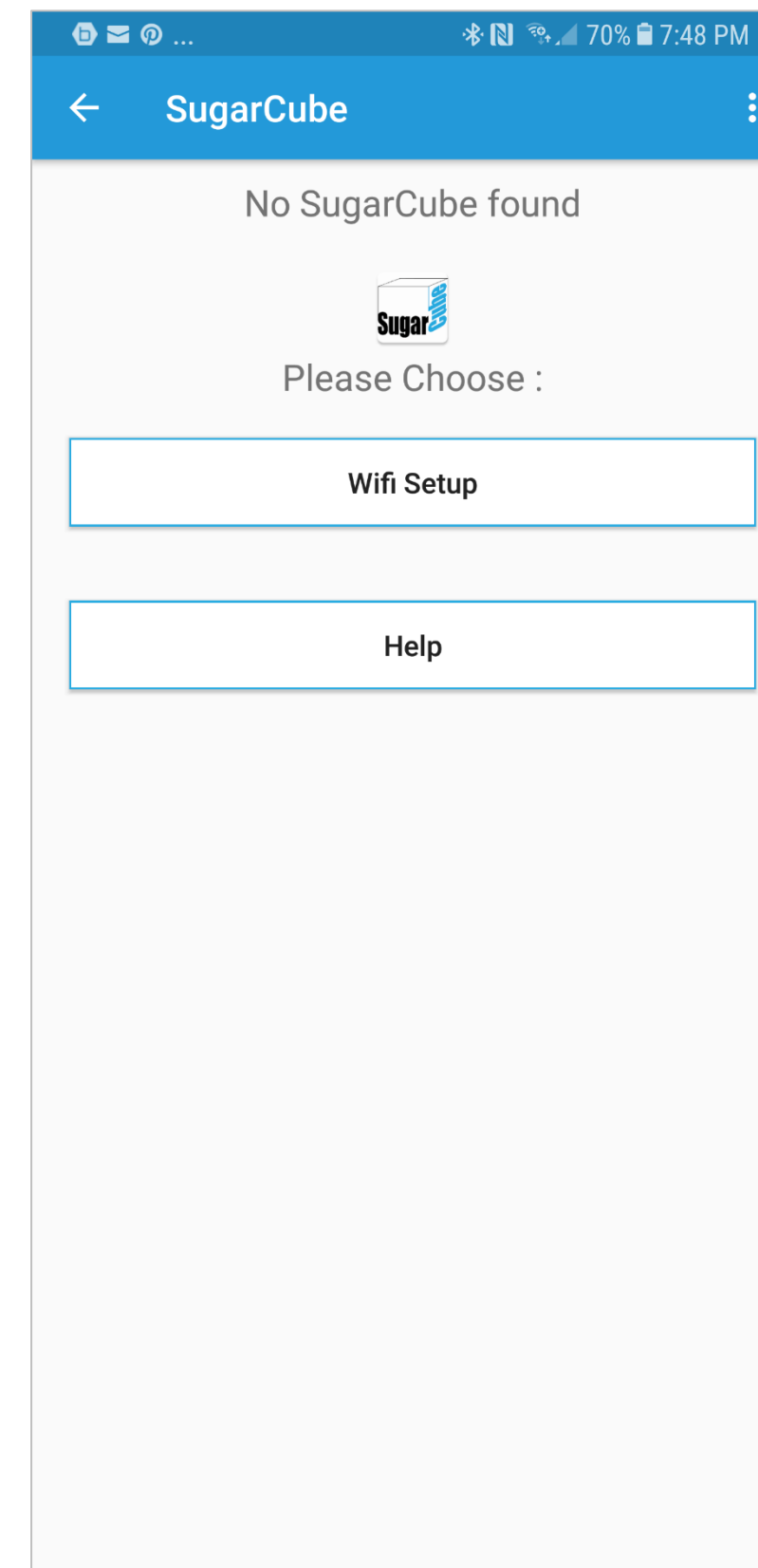
33. Connecting the app with SugarCube on Wi-Fi



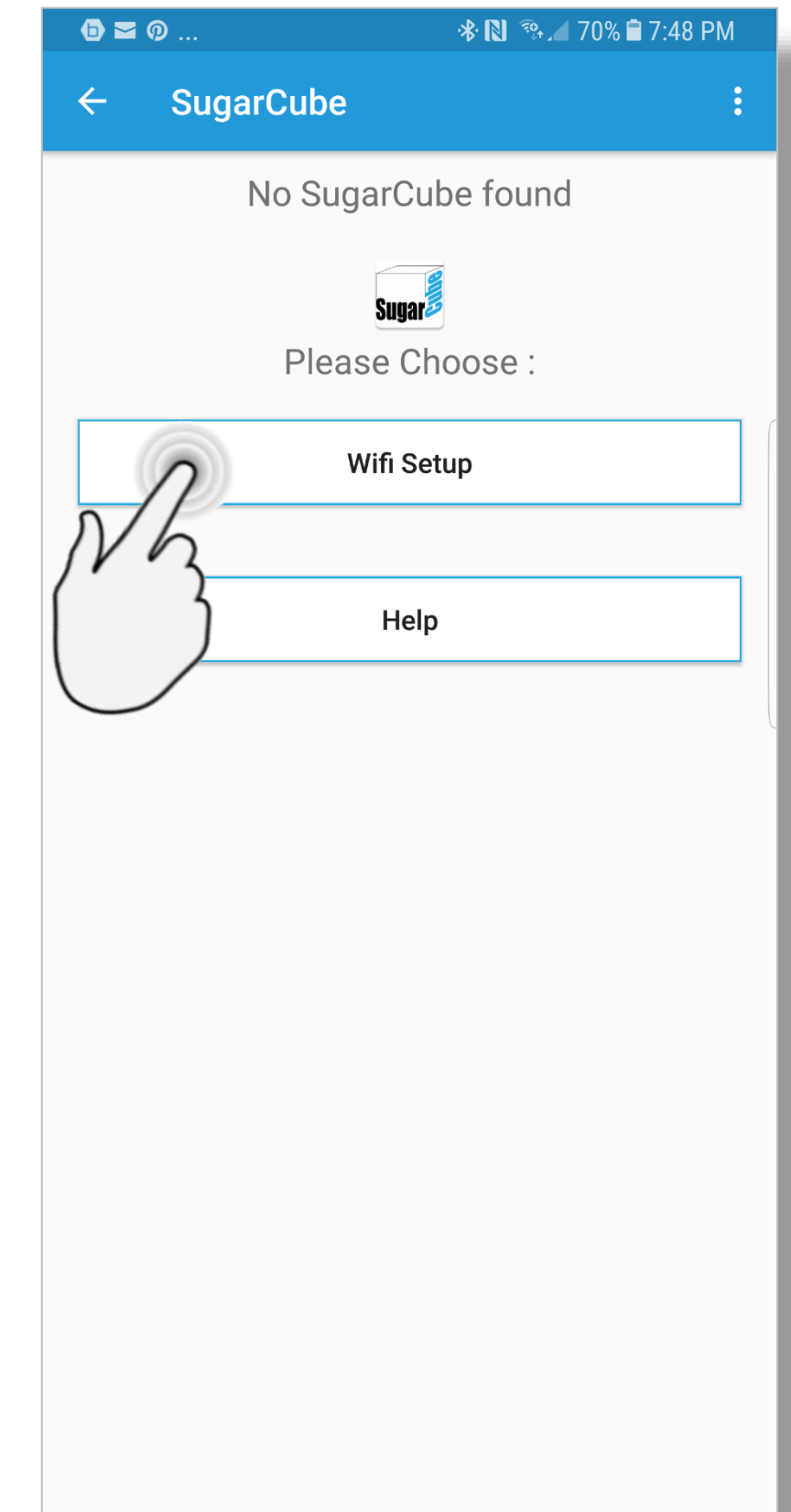
(1) App is loading



(2) App is searching for SugarCubes on the LAN

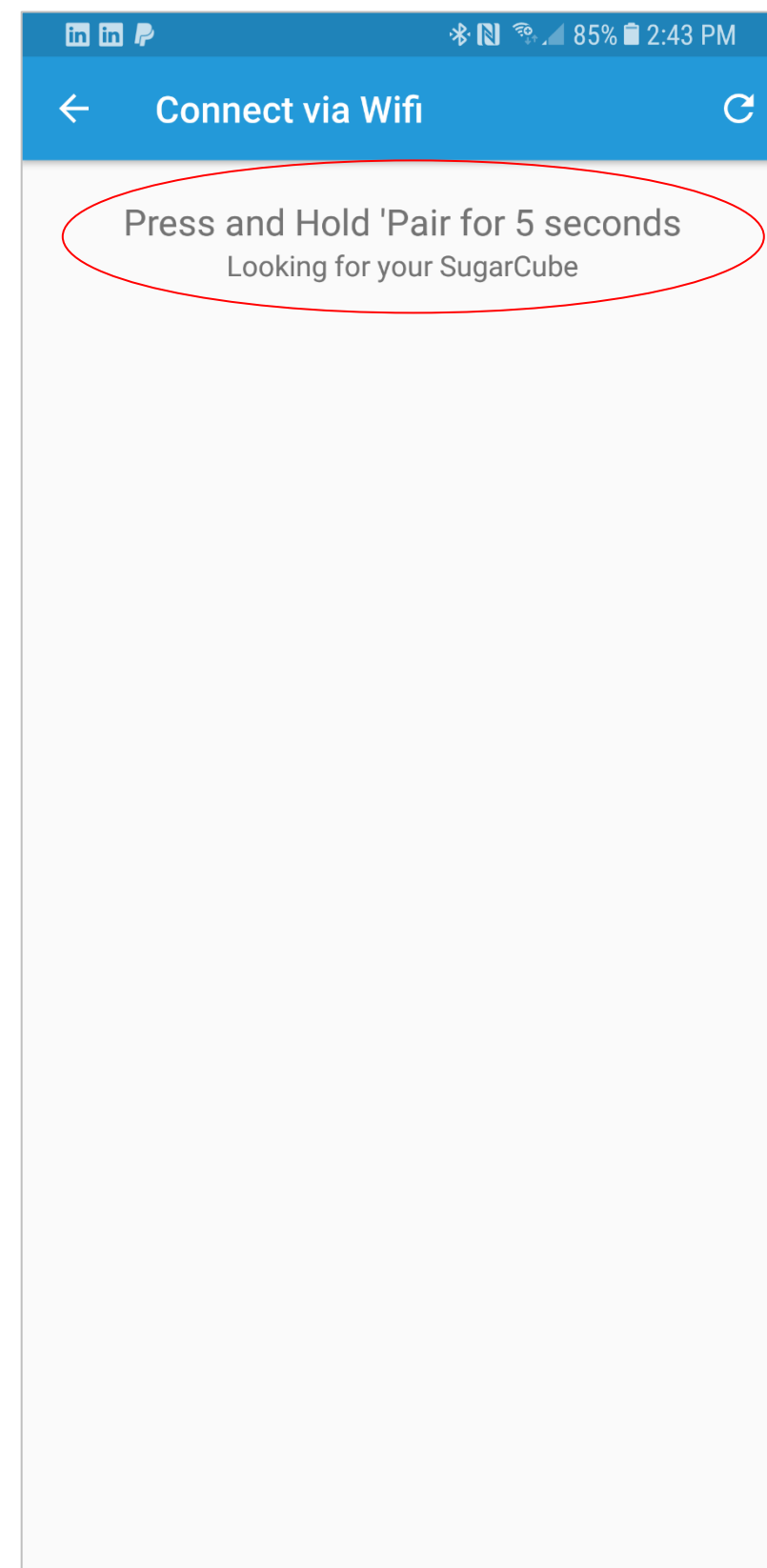


(3) No SugarCube is found so the app offers the option to run WiFi setup.

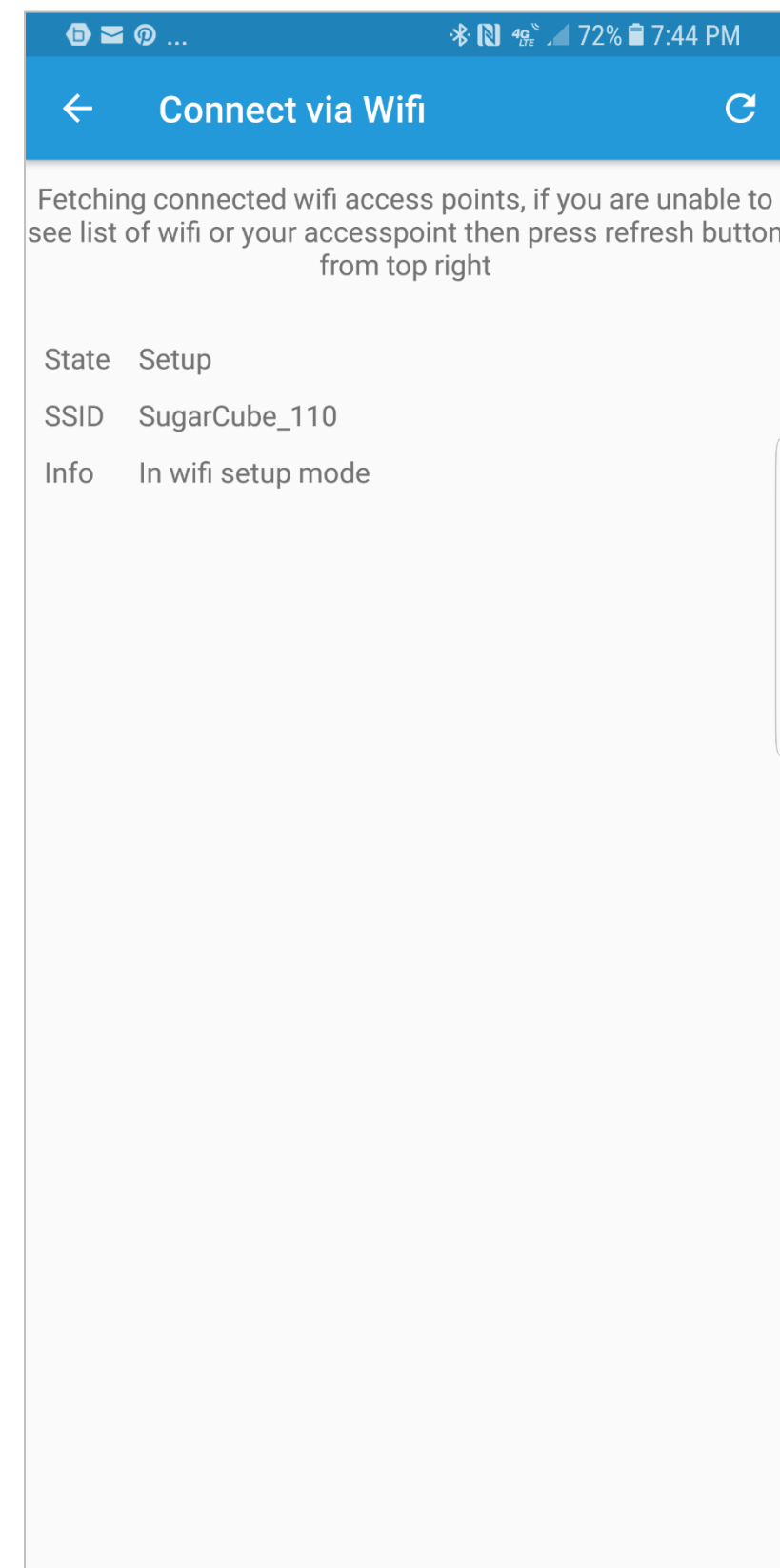


(4) User taps 'WiFi Setup' to start the process.

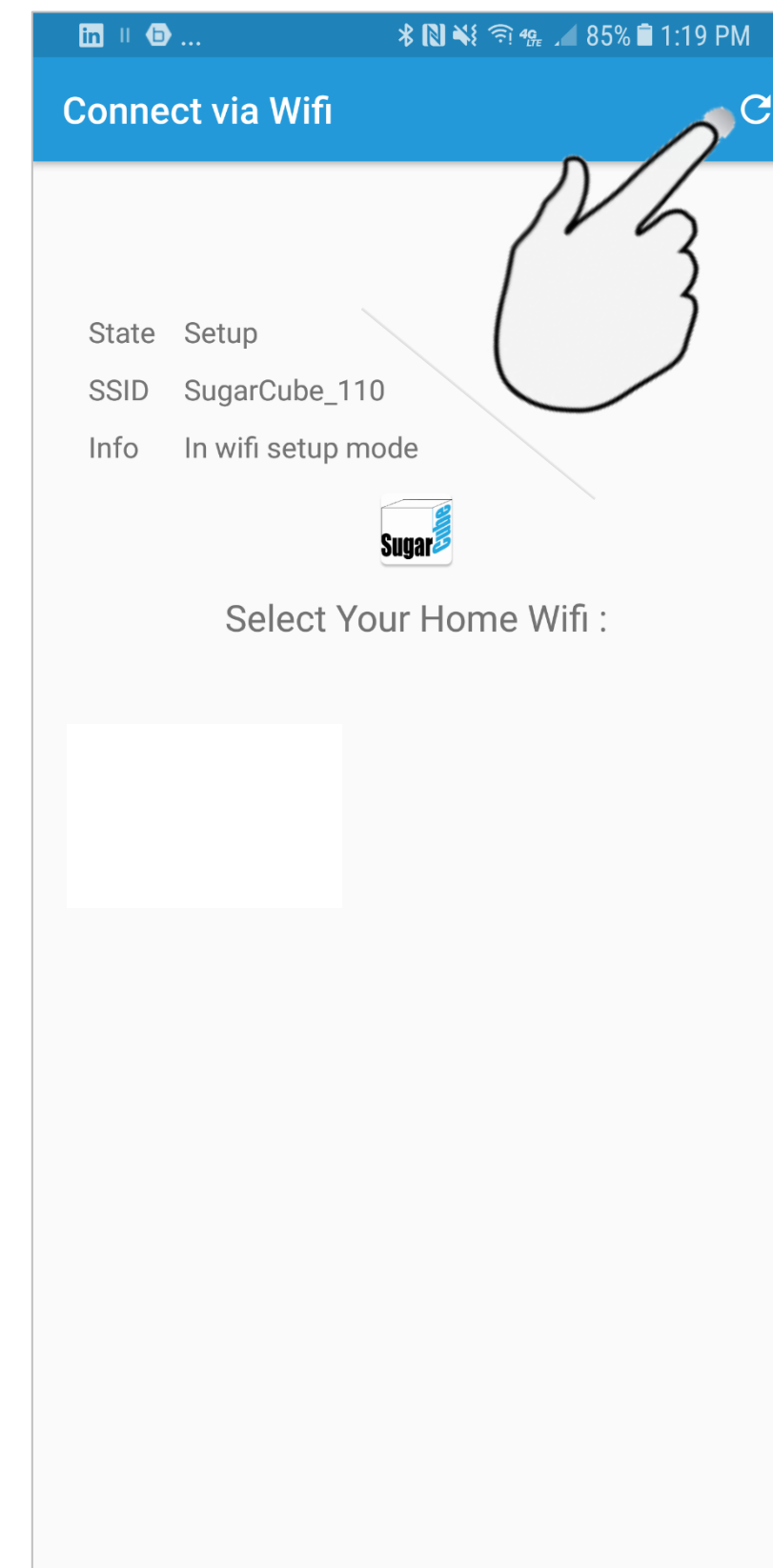
34. Connecting the app with SugarCube on WiFi



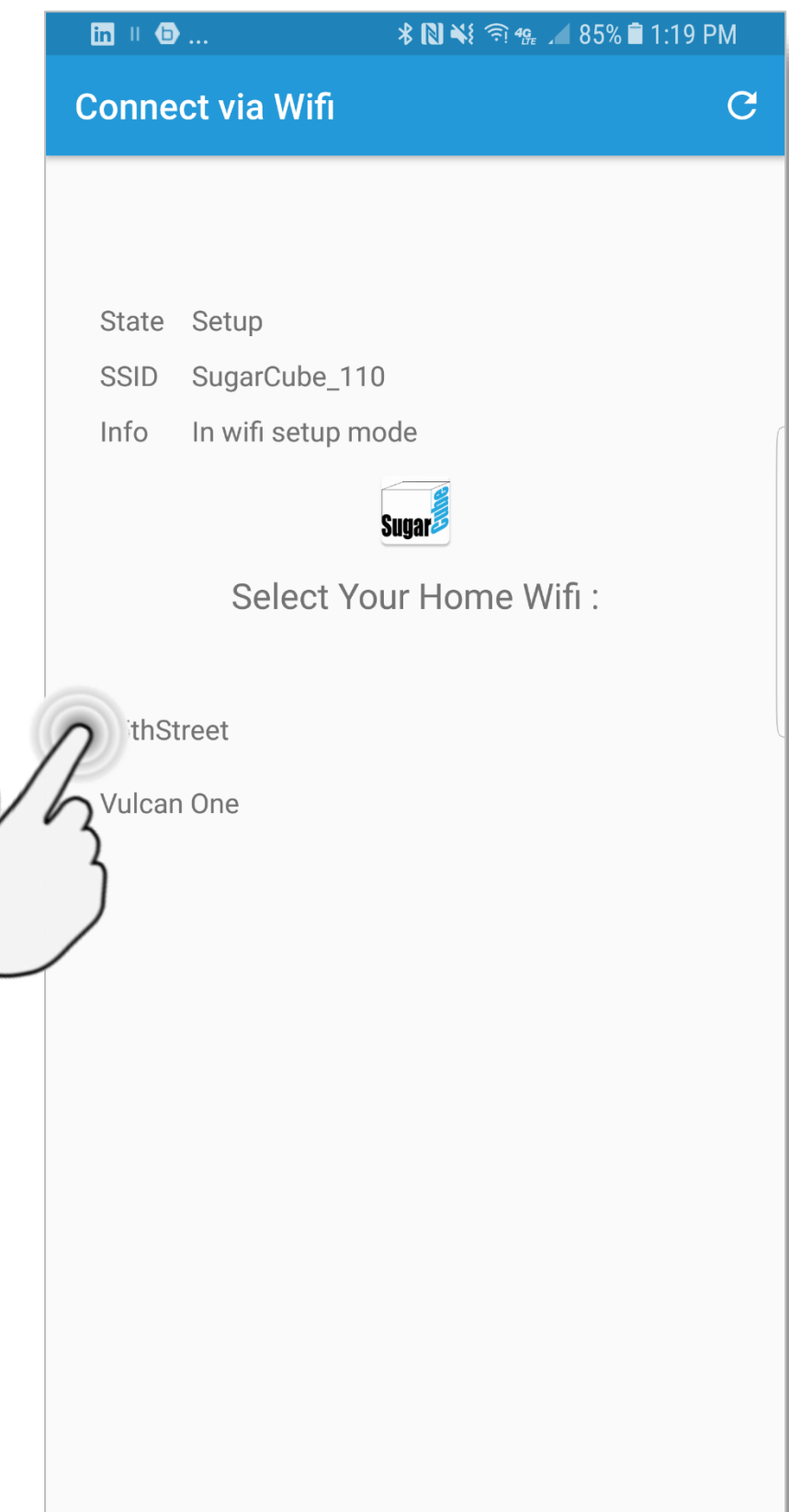
(5) Next, press and hold the pairing button (on the rear panel) for 5 seconds as instructed.



(6) After this, your SugarCube will be configured as an access point and display its SSID (e.g. SugarCube_110). In this screen the app has now connected to it automatically. The password is: **SugarCube**

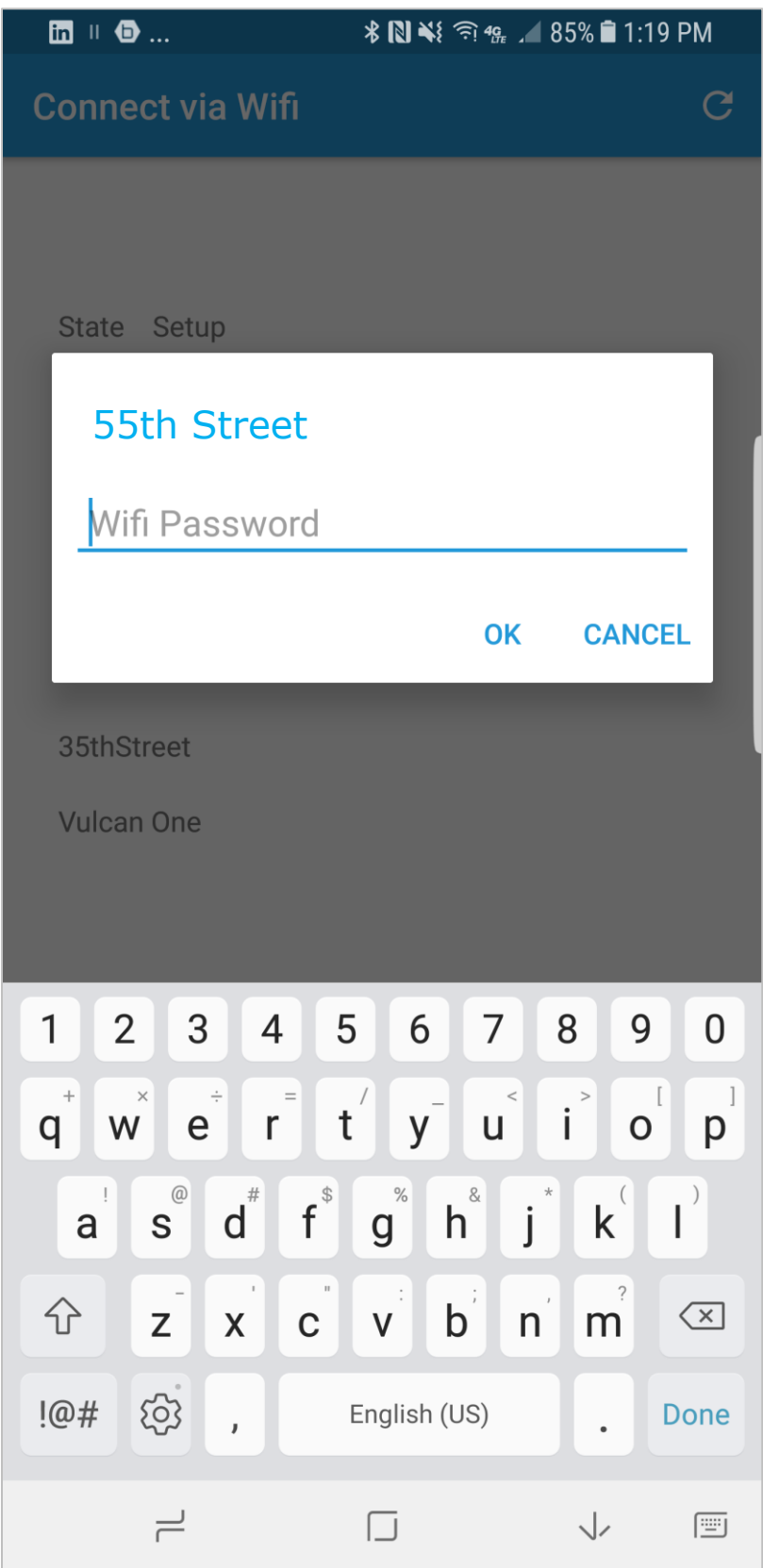


(7) Next, the SugarCube will find your available WiFi connections. This could take a minute but if nothing shows, please press the refresh (upper right).

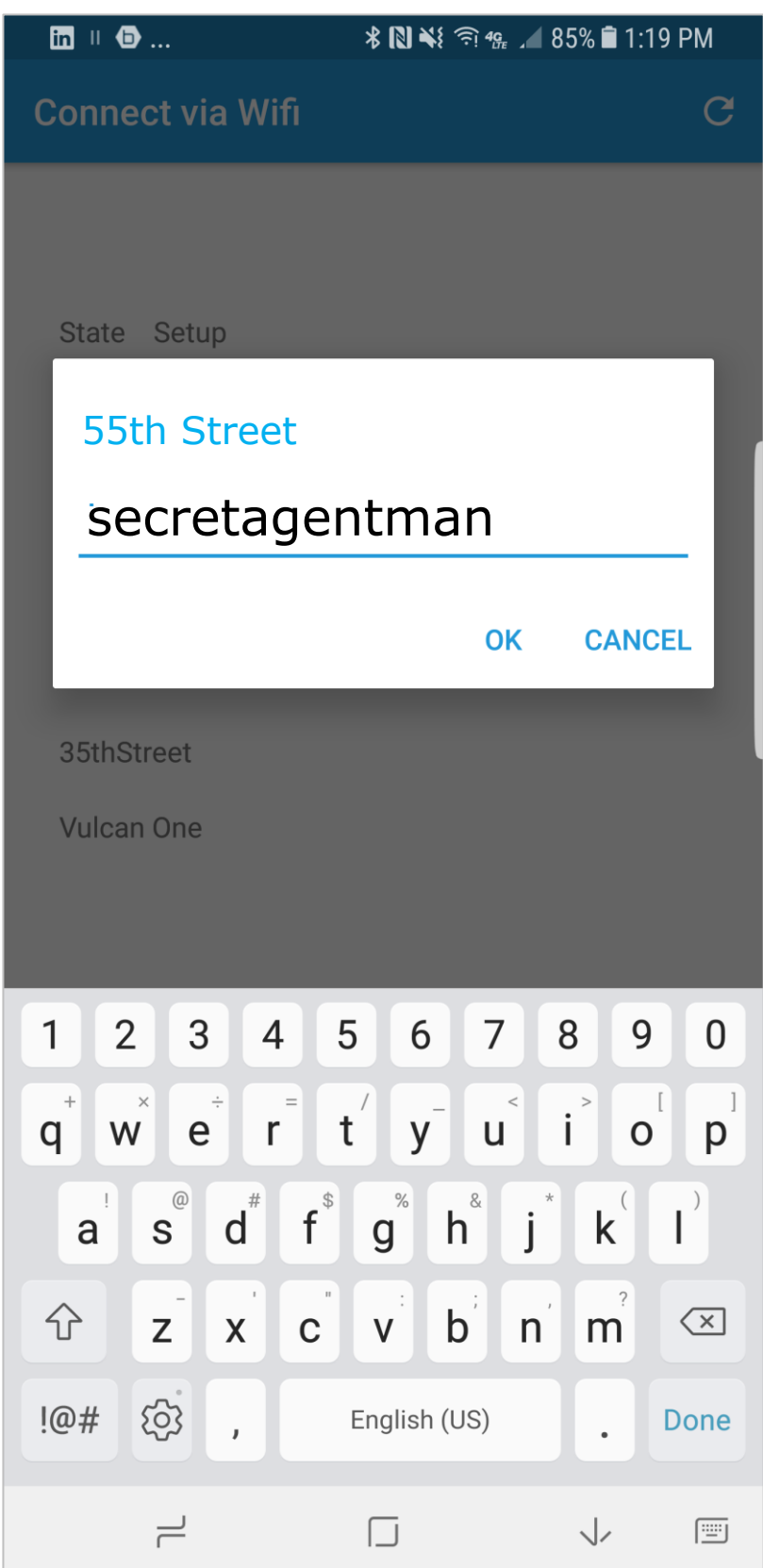


(8) When you see your network, select the desired connection.

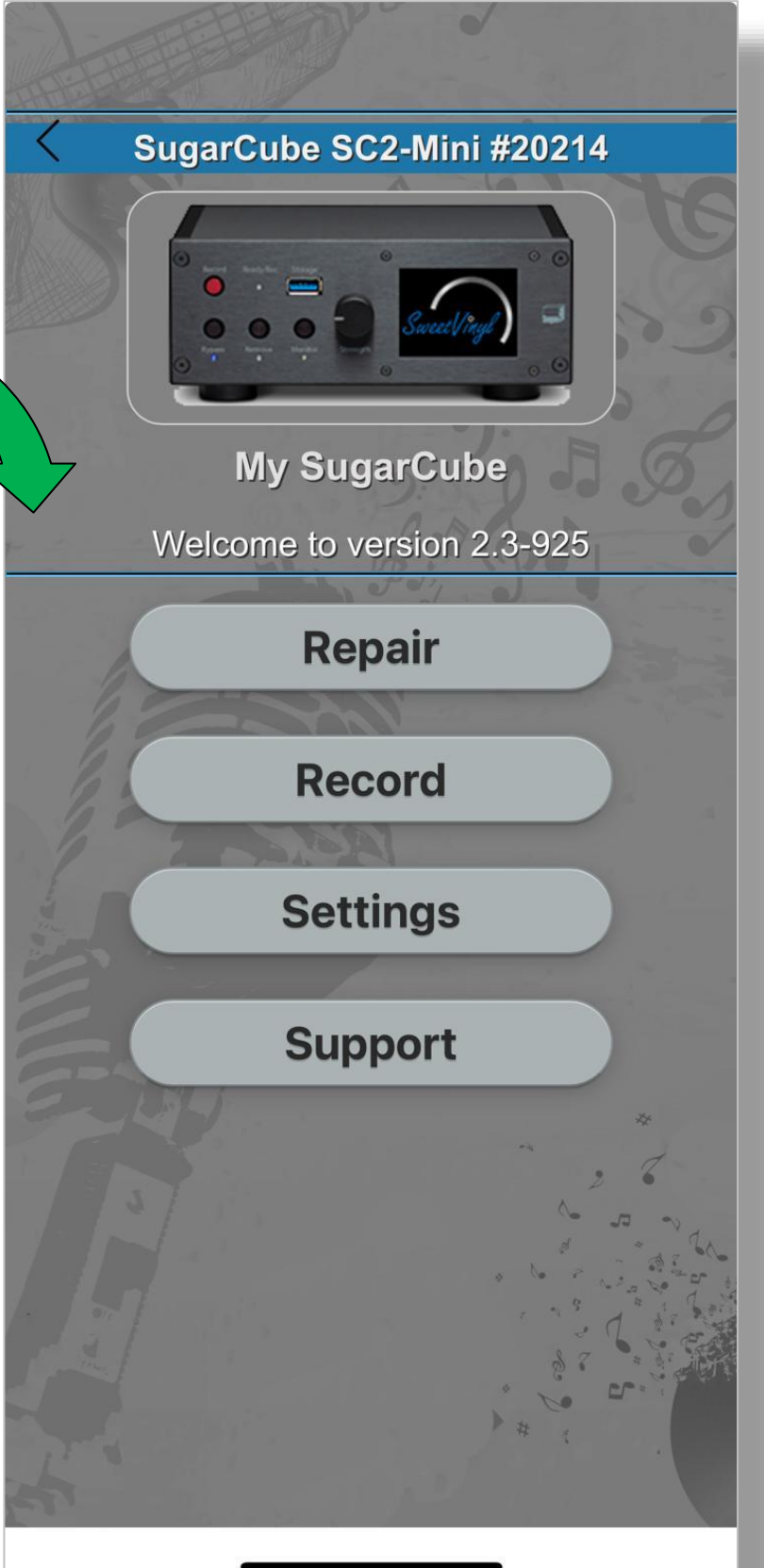
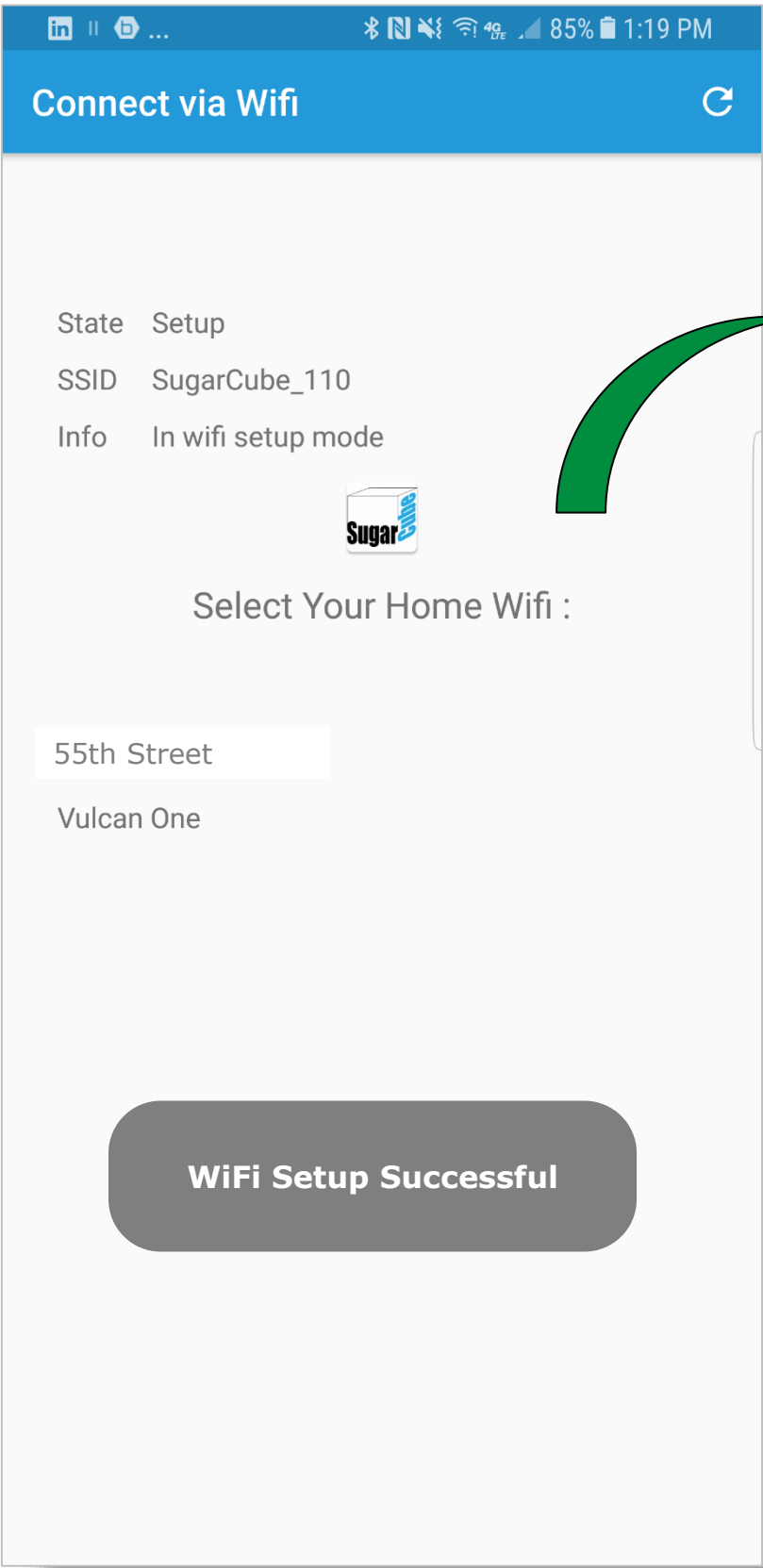
35. Connecting the app with SugarCube on Wi-Fi



(9) Next, enter the password for your WiFi network.



(10) The app has connected successfully with the SugarCube and will now jump to the main screen.



(11) The connection process is complete and you are ready to go.



Using **Wired Ethernet** with SugarCube

36. Connecting Your SC2-Mini via Wired Ethernet

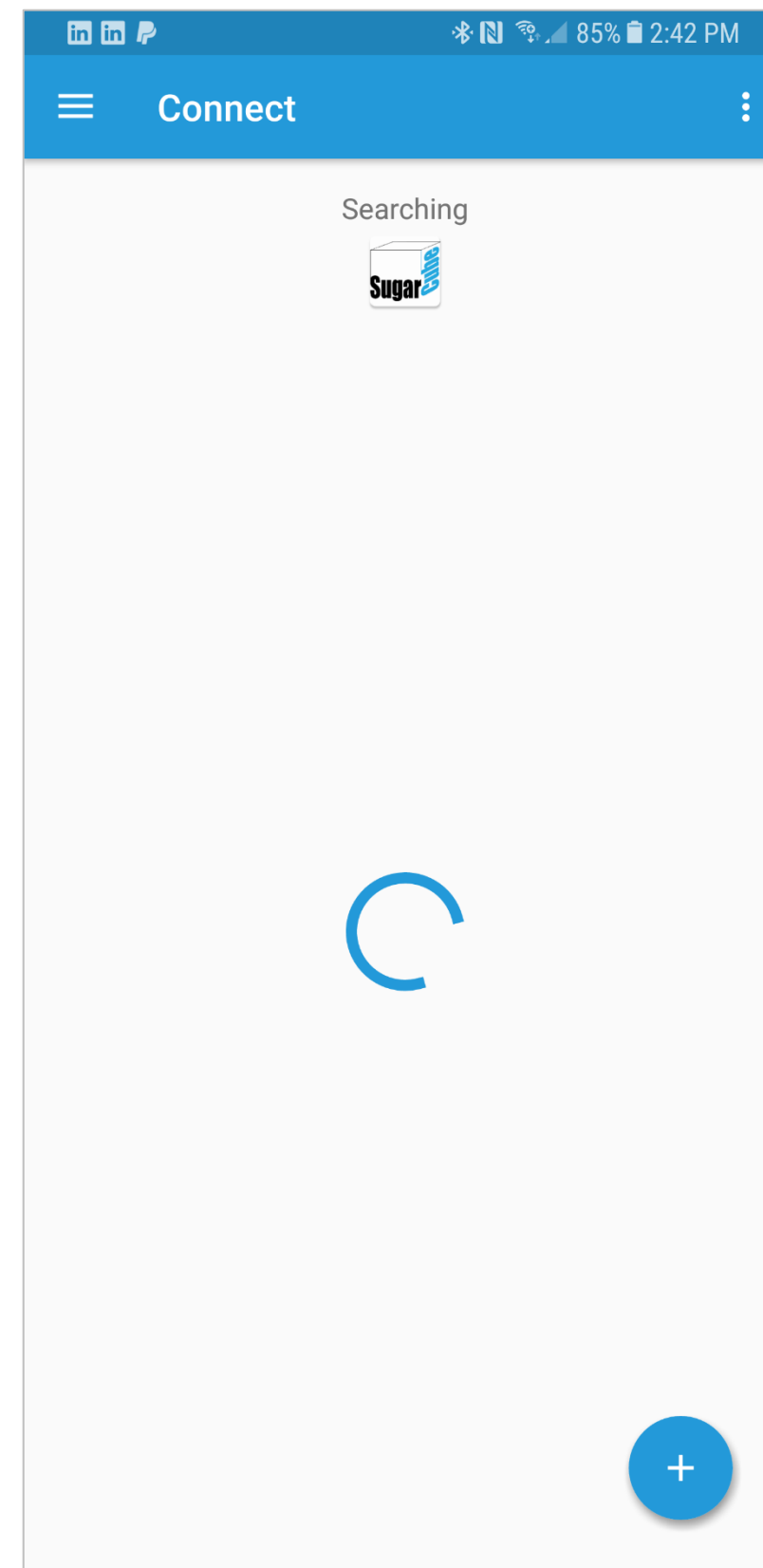


If you would like to use the SugarCube app to control your SugarCube over wired Ethernet, plug in the cable to the back panel as seen above and follow the steps on the next few pages.

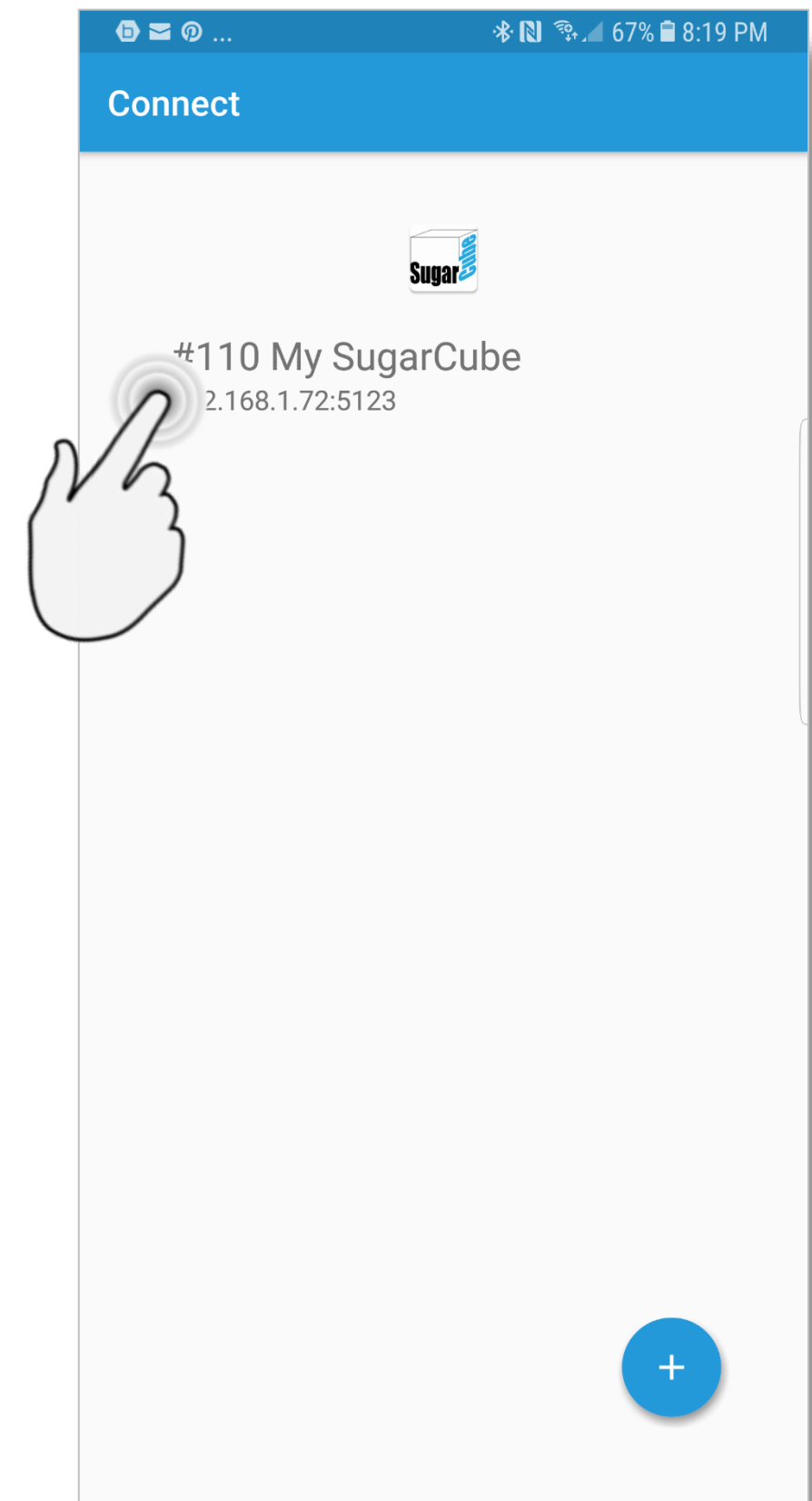
37. Connecting the app with SugarCube on wired Ethernet



(1) App is loading

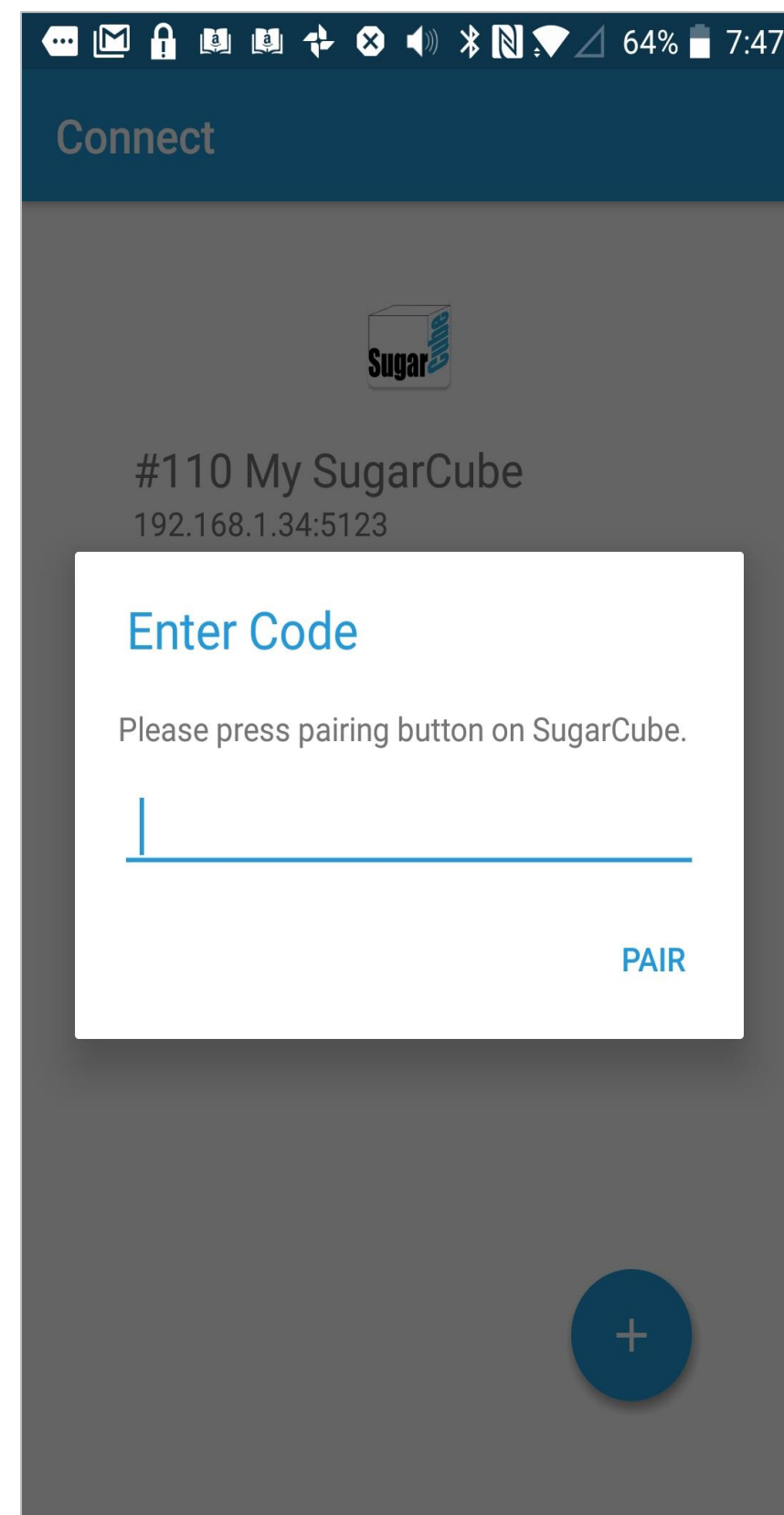


(2) App is searching for SugarCubes on the LAN

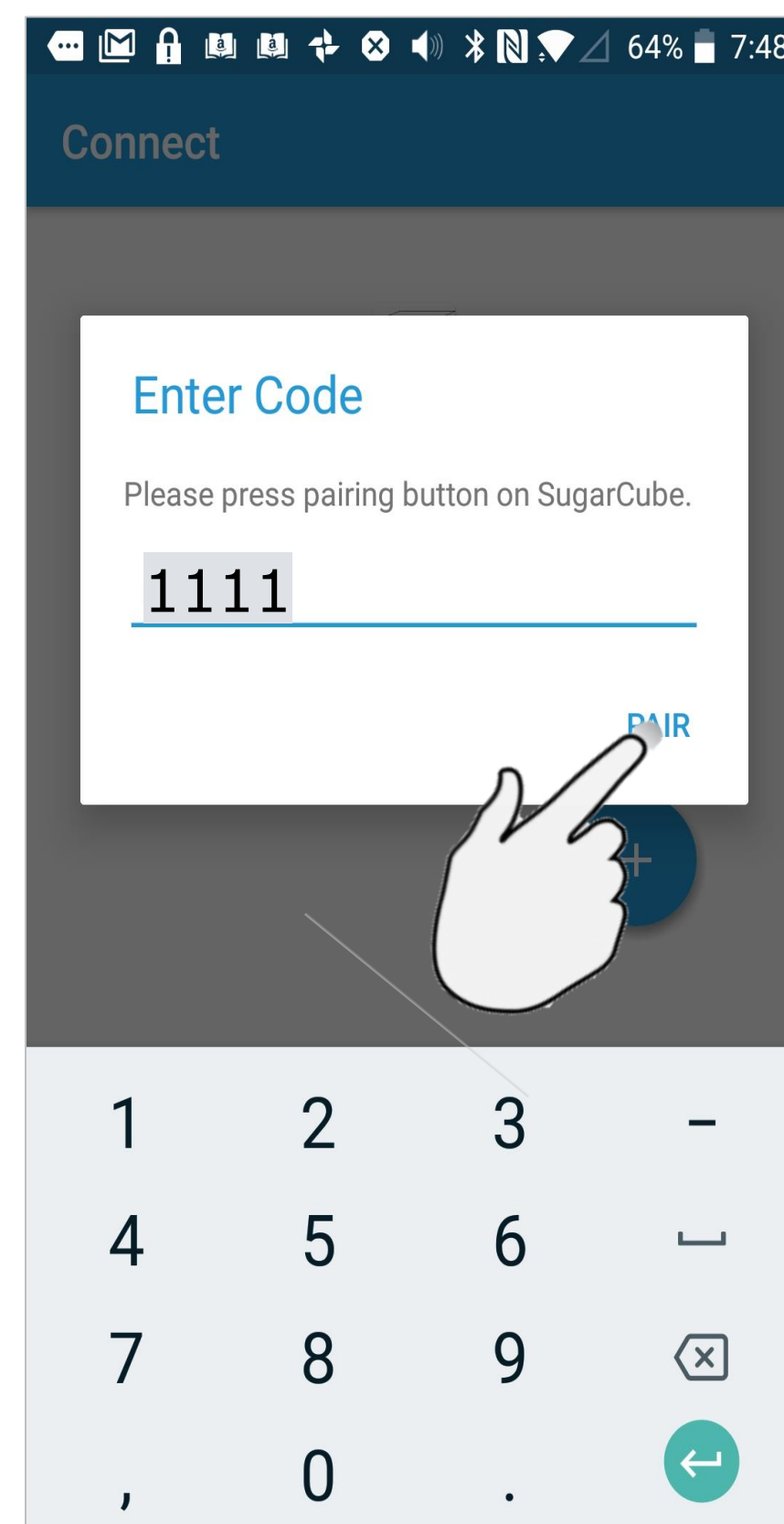


(3) App has found a SugarCube and will now initiate a connection.

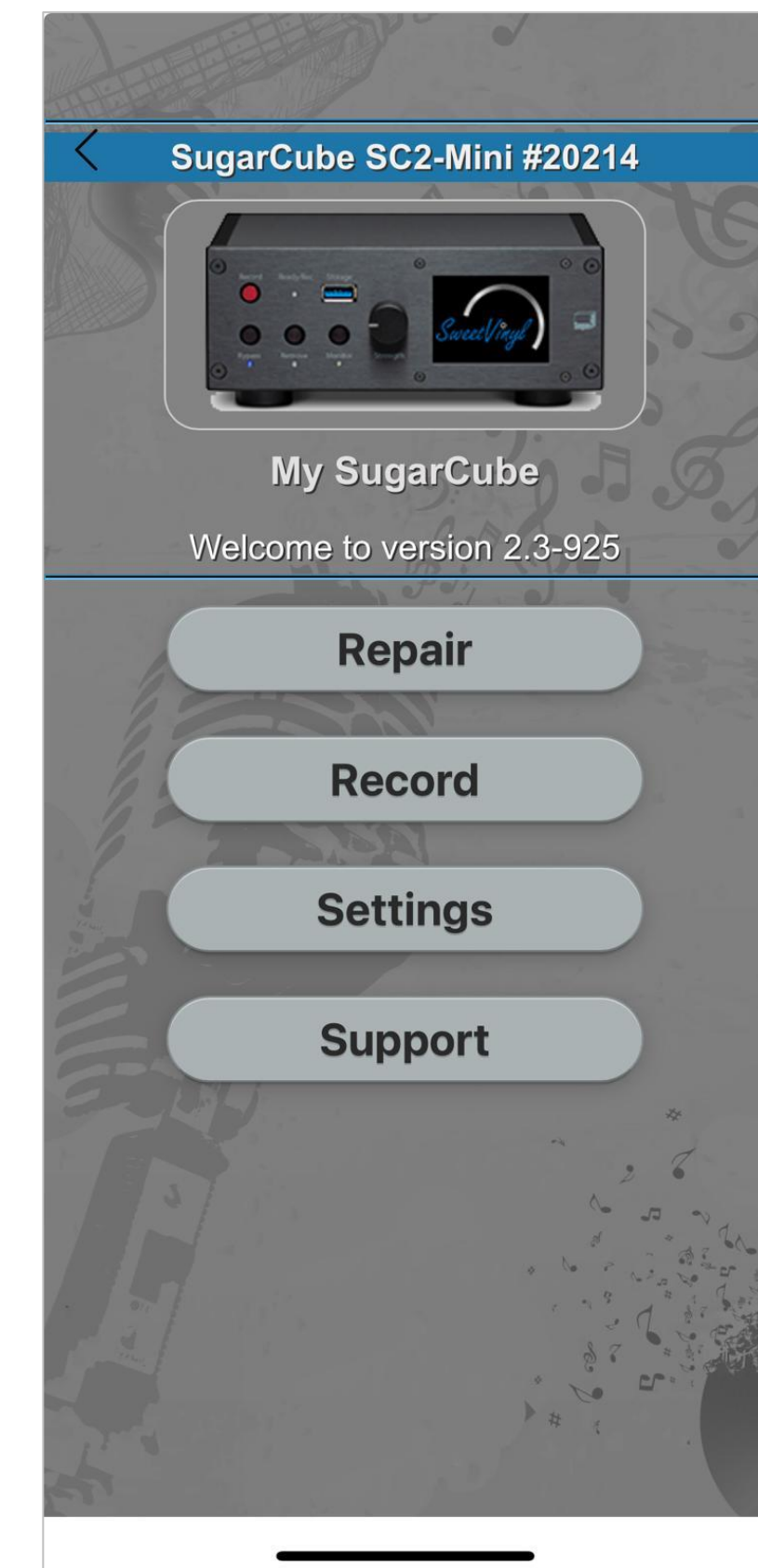
38. Connecting the app with SugarCube on wired Ethernet



(4) If you have more than one SugarCube, press the front panel 'Sensitivity' knob to get a pairing code. Otherwise, just enter '1111'.



(5) For a single SugarCube, enter '1111'



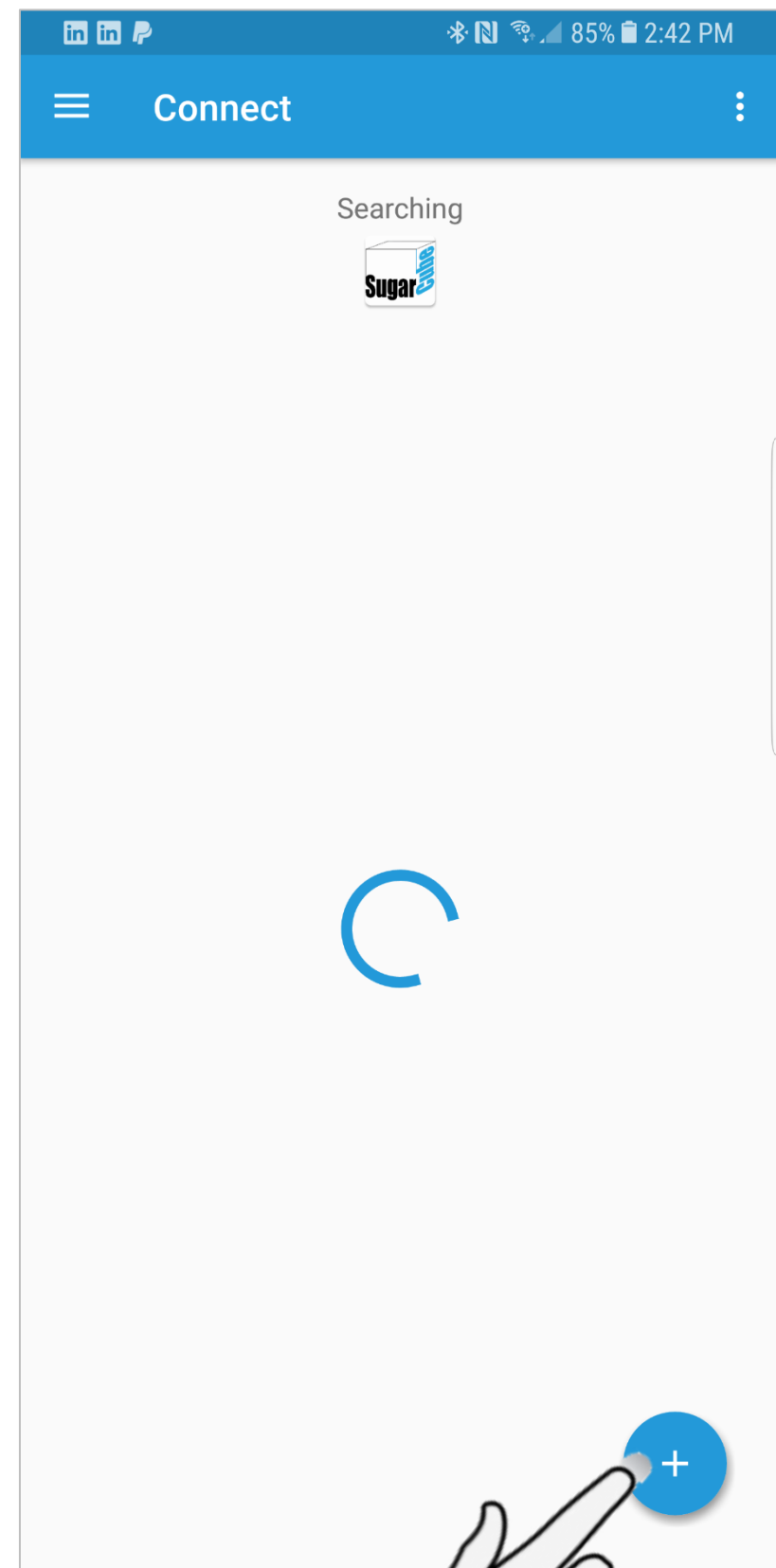
(6) App is linked to SugarCube and ready for use.



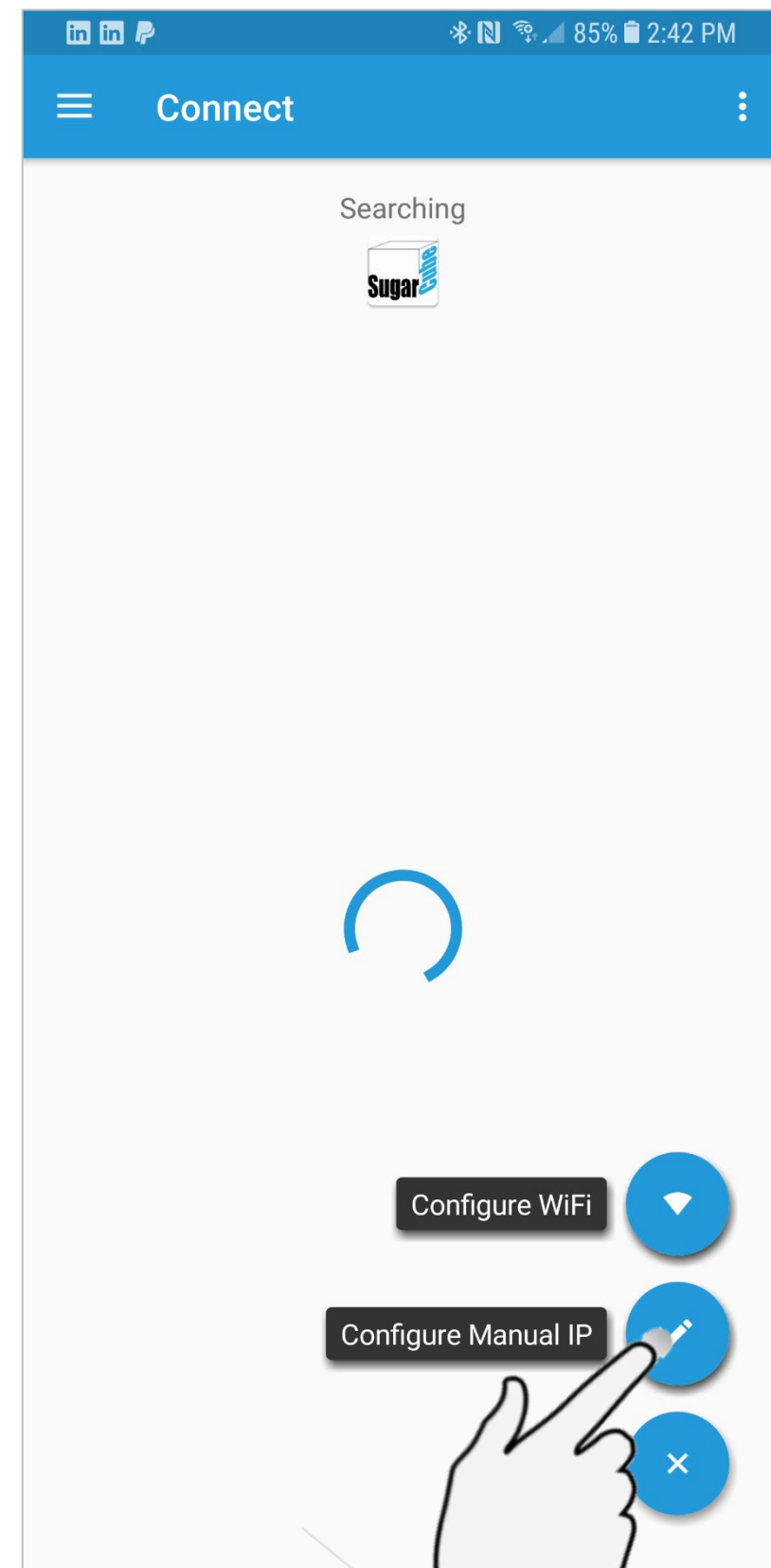
Manually Connecting to SugarCube

Using the SugarCube's IP Address

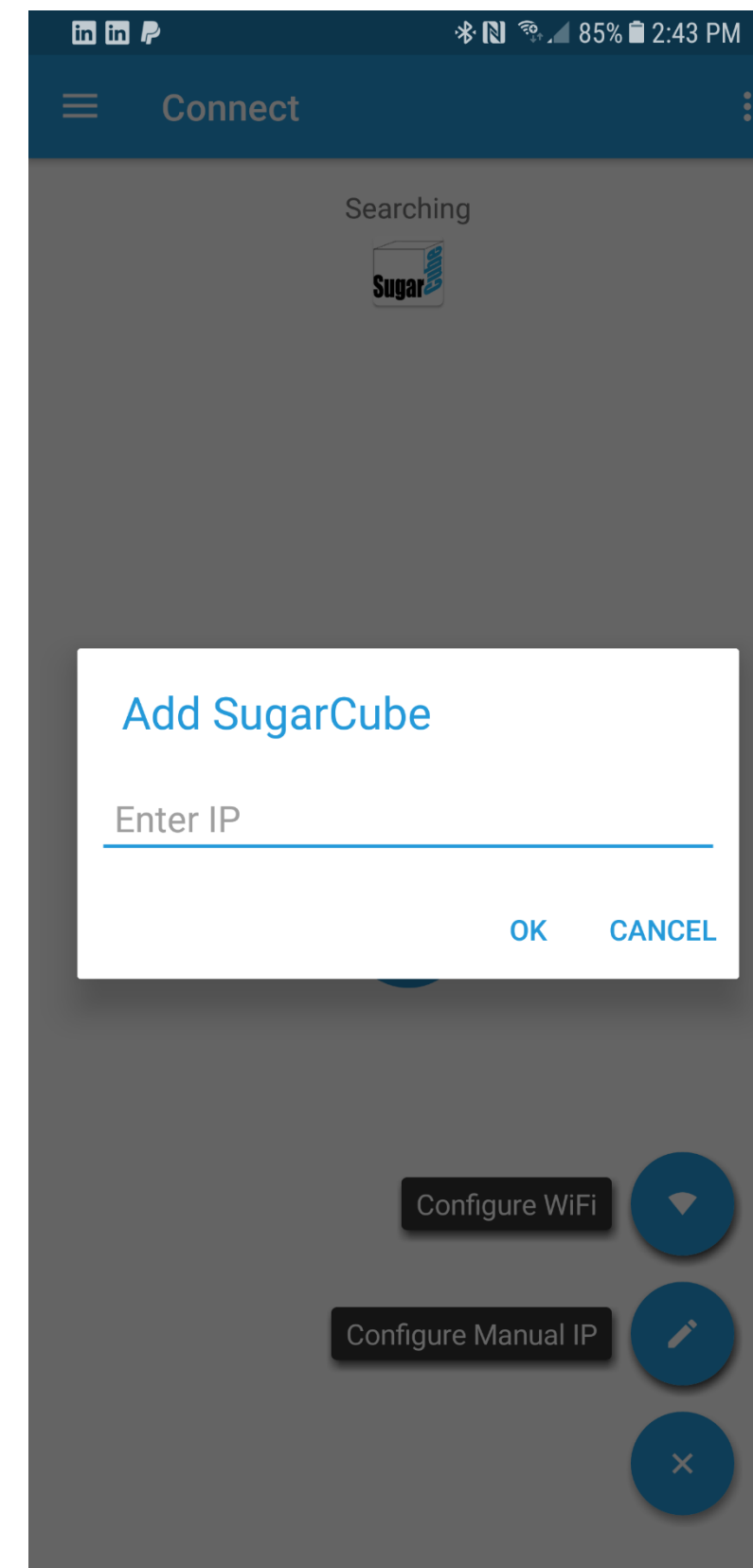
39. Manually Connecting the App Using IP Address



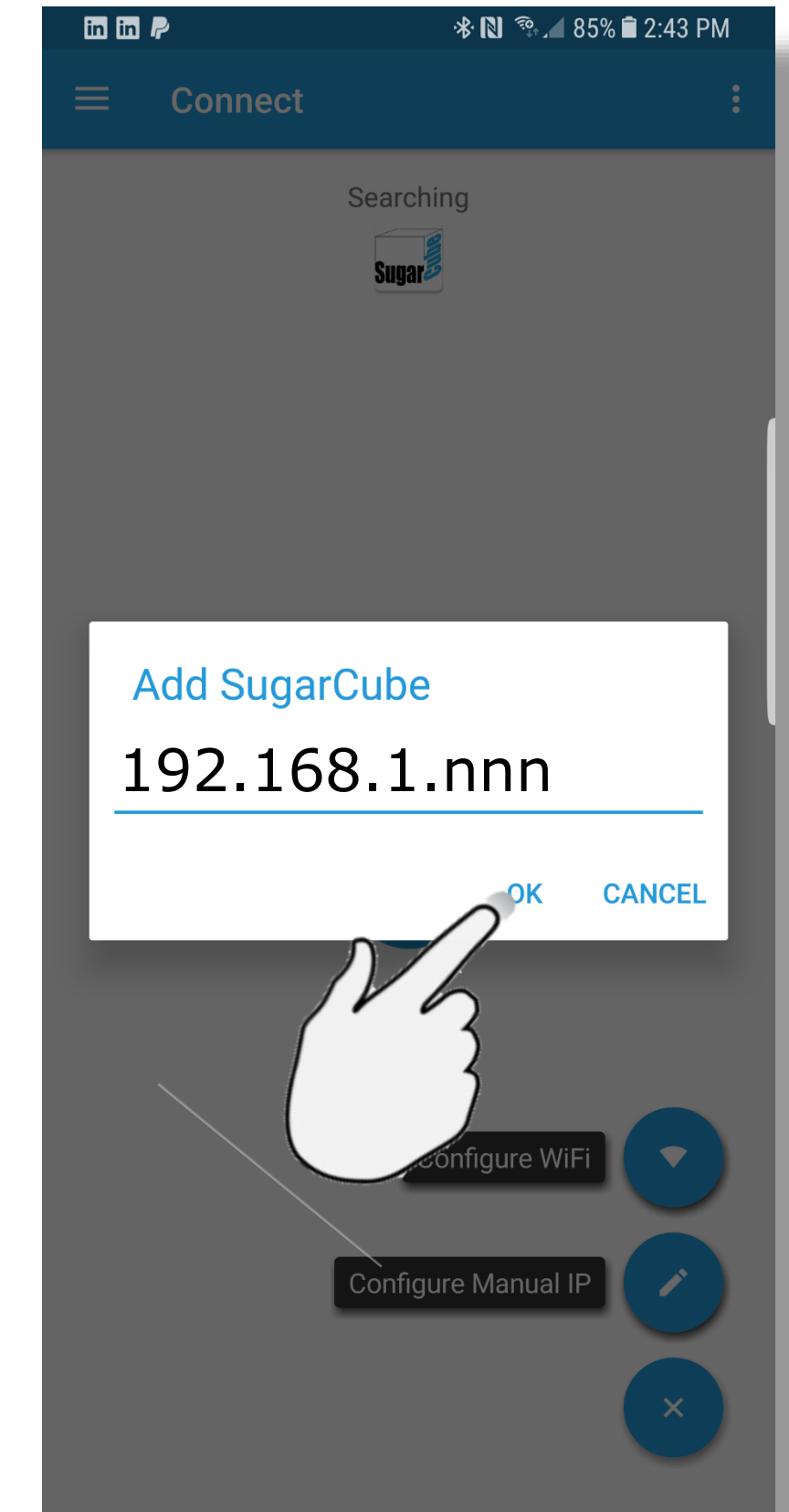
(1) If, for any reason, you wish to manually connect to your SugarCube touch the menu button in the lower right of the screen.



(2) Select 'Configure Manual IP'.

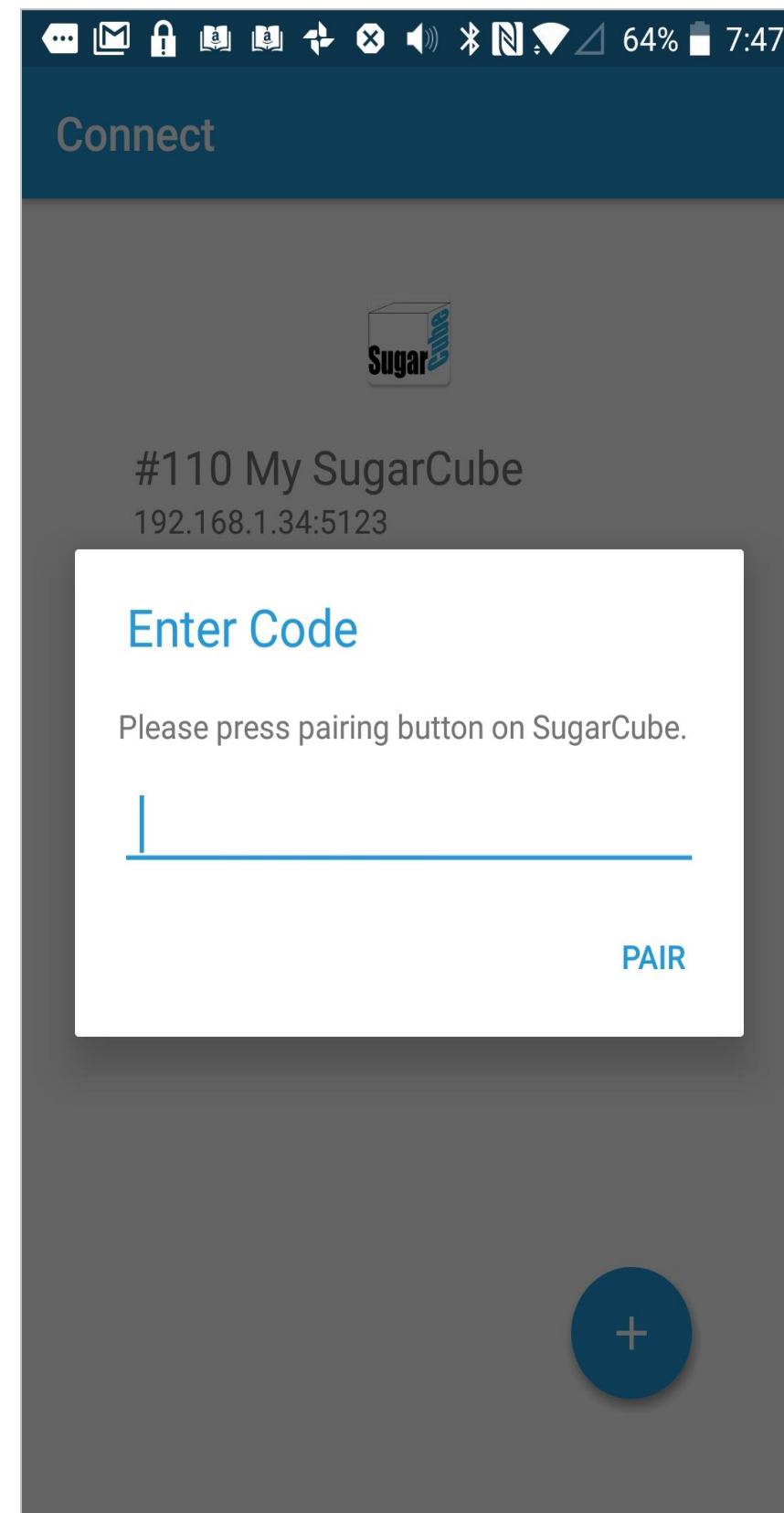


(3) You will then see a window to enter the IP address of the SugarCube. This address can be seen by pressing and holding 'Bypass' for a few seconds.

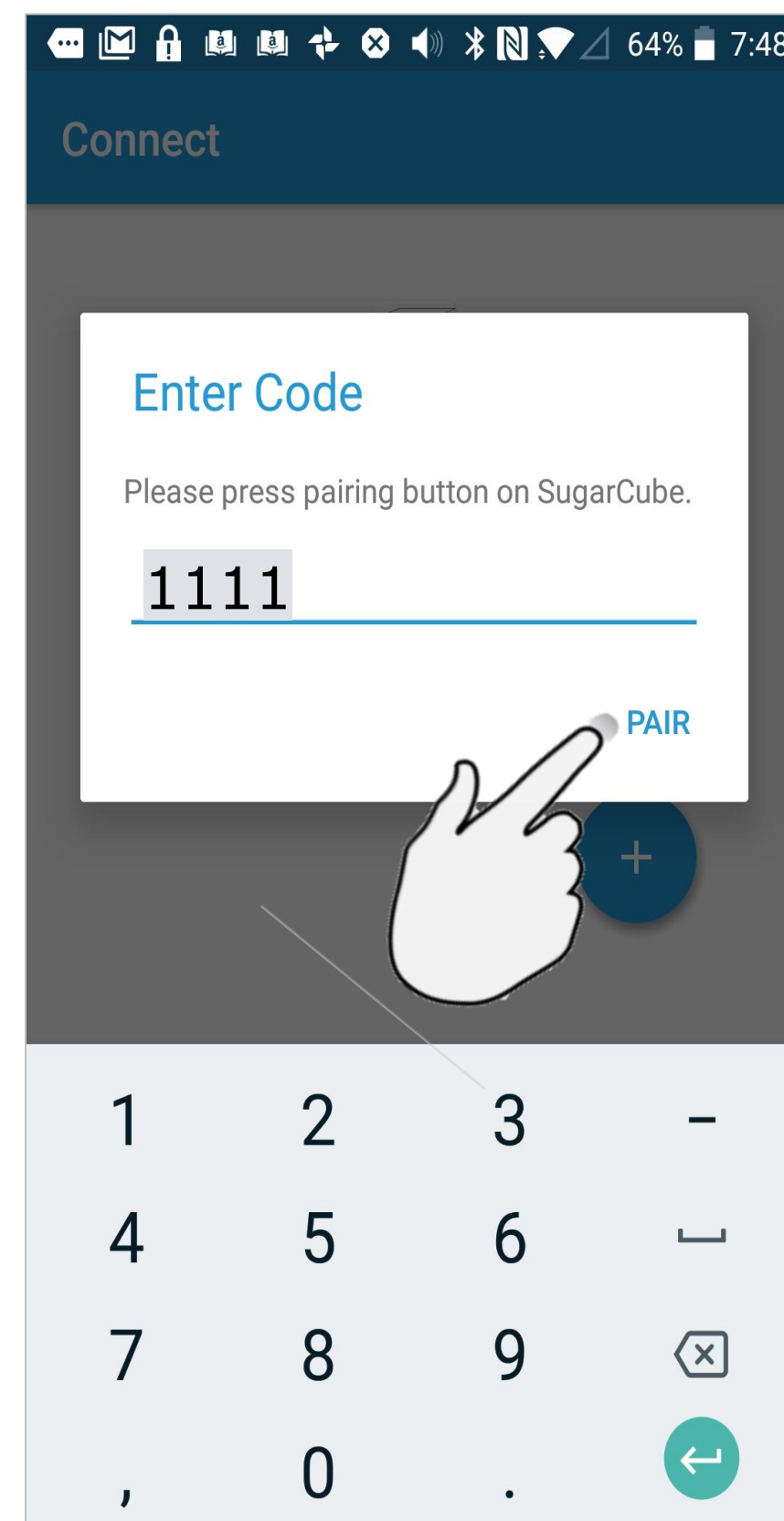


(4) Enter the IP address as displayed on the front panel display.

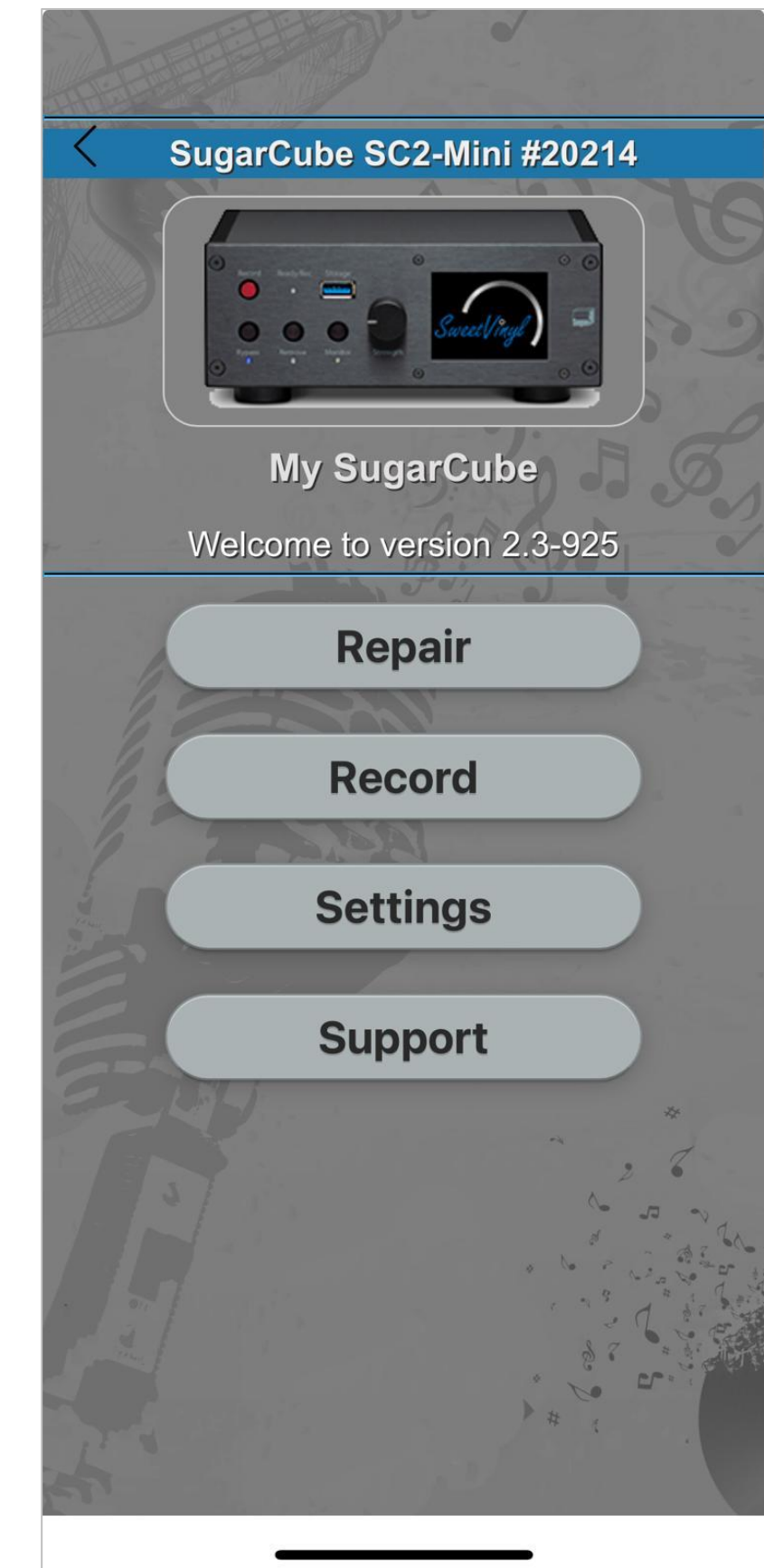
40. Manually Connecting the App Using IP Address



(5) If you have more than one SugarCube, press the front panel 'Sensitivity' knob to get a pairing code. Otherwise, just enter '1111'.



(6) For a single SugarCube, enter '1111'



(7) App is linked to SugarCube and ready for use.

Technical Specs

Appendix A

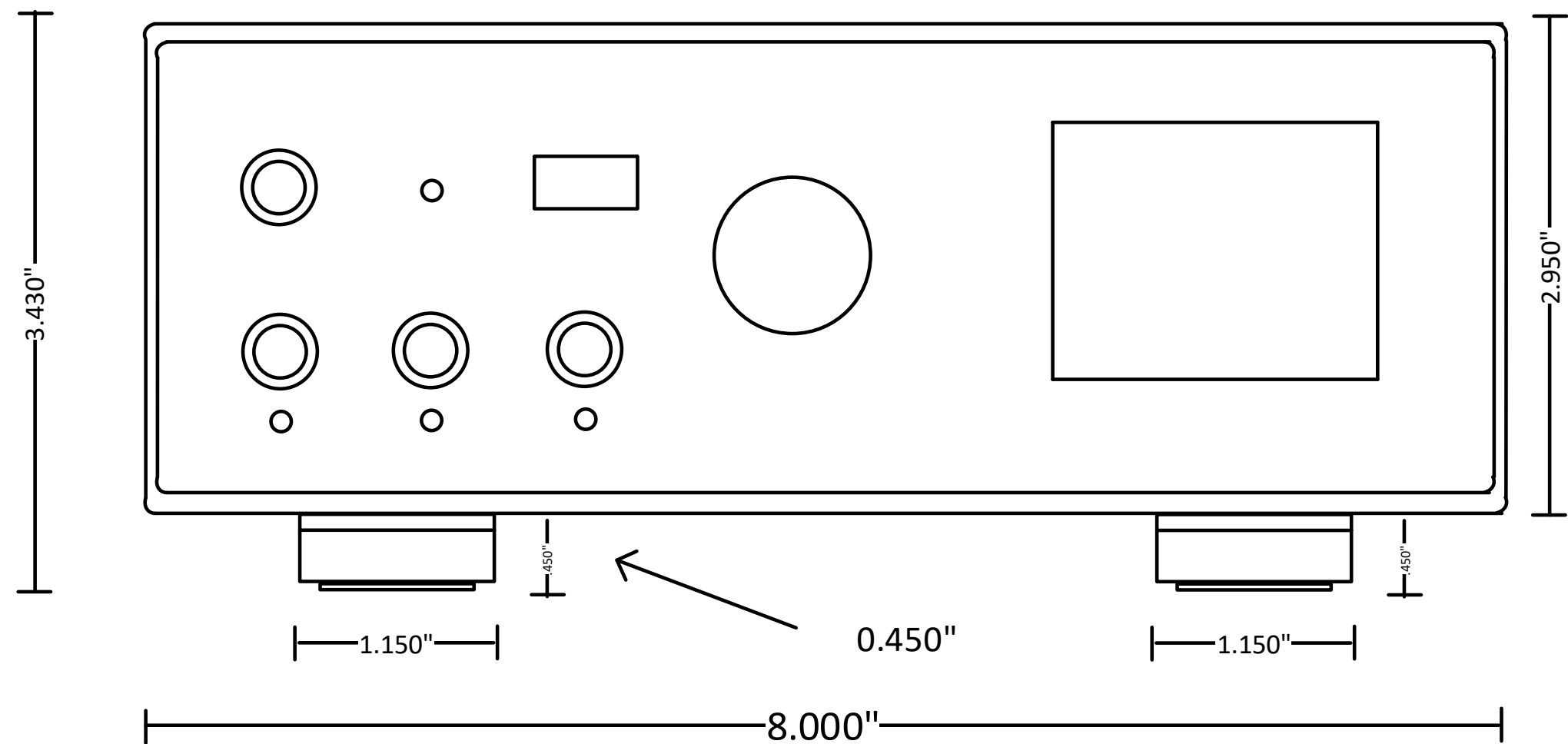


SugarCube SC2-Mini Mechanical:

SC-2 Mini
Side View



Front View



Technical Specifications

- **Line In** maximum level = +20 dBu (approx. 7.7Vrms or 22Vp-p)
- **Line In** impedance = 100k Ohm
- **Line Out** level always equal to **Line In** level, regardless of any internal settings
- THD+N = 0.0030% (A-weighted)
- Frequency Response = 5Hz to 75kHz -3dB
- DC in = 5 volts, center-pin positive
- Average Power Dissipation = 8 watts
- Internal digital process: 192kHz/32bits
- External digital file sample rates: 192k to 44.1k, 16 or 24bit
- External digital file formats: WAV, AIFF, FLAC, ALAC, AAC, MP3

Support

Thank you for purchasing this SugarCube product.

Please contact customer support at: support@sweetvinyl.com

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Compliance

For the current EU Declaration of conformity, visit:
<http://support.sweetvinyl.com/regulatory>

For regulatory compliance information in North America, visit:
<http://support.sweetvinyl.com/regulatory>

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